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MILITARY HANDBOOK

NUCLEAR WEAPONS SYSTEMS, SAFETY, DESIGN, AND EVALUATION CRITERIA FOR

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DEPARTMENT OF THE NAVY
WASHINGTON, DC 20360

MIL-HDBK-255A(AS)

NUCLEAR WEAPON SYSTEMS SAFETY, DESIGN
AND EVALUATION CRITERIA FOR

1. This handbook, prepared by the Naval Air Systems Command, was approved 5 May 1978, for printing and inclusion in the Military Standardization Handbook series.

2. This document provides basic and fundamental information on nuclear safety design and evaluation criteria. It will provide valuable information on nuclear safety design and evaluation criteria. It will provide valuable information and guidance to personnel concerned with the design and development of nuclear weapon systems and the evaluation criteria that is used to ensure nuclear safety.

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Existing Department of Defense and Department of the Navy implementing instructions for nuclear weapon systems safety establish design requirements and evaluation criteria which are general in nature. These requirements must be analyzed to determine suitability and applicability for a given system development and procurement.

This document provides additional information and guidance to the program manager, acquisition engineer, and others responsible for the design, test, and procurement of nuclear weapons components, equipment, subsystems, and systems on the nuclear safety design and evaluation criteria at various stages of their life cycle. The intent is not to provide detailed nuclear safety design and evaluation criteria, but rather to provide guidance and information which should be considered during design and procurement.

Every effort was made to assure that the referenced documents (Section 2) were in effect on the date of publishing this document. However, the currency of each document should be checked before use.

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SECTION 1
INTRODUCTION

1.1 SCOPE. The intent of this handbook is to provide nuclear safety design and evaluation criteria which apply to all nuclear weapon systems for which funding and tasking are the responsibility of the Naval Air Systems Command (NAVAIRSYSCOM). The nuclear safety standards of the Department of Defense are the evaluation base. In addition, the malfunction or accidental operation of any single component must not result in the prearming, arming, launching, firing, or releasing of a nuclear weapon.

1.2 PURPOSE. This handbook provides information on the nuclear safety design and evaluation criteria to be considered by program managers (PMs), acquisition managers (AMs), and NAVAIR contractors responsible for the design and procurement of new systems, subsystems, and equipment which may be associated with nuclear weapons or warheads during their life cycle.

a. This handbook should not be interpreted as a requirement to modify existing operational systems which do not meet the safety criteria outlined herein. However, when an existing system is modified, the system must be evaluated to determine the extent to which these criteria should apply. As a minimum, the criteria shall be considered for that part of the system being modified.

b. This handbook does not apply to compatibility certification, but such certification is required before delivery configurations involving nuclear weapons/warheads can become operational.

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SECTION 2
REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this handbook to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-H-904	Hoists, Chain, Hand Operated, Hook and Trolley Suspension
MIL-B-5087	Bonding, Electrical and Lightning Protection, for Aerospace Systems
MIL-E-6051	Electromagnetic Compatibility Requirements for Systems
MIL-M-8090	General Requirements for Mobility Towed Aerospace Ground Equipment
MIL-A-8421	General Specifications for Air Transportability Requirements
MIL-S-8512	General Specifications for the Design of Support Equipment, Aeronautical, Special
MIL-I-8671	Installation of Droppable Stores and Associated Release Systems
MIL-H-19925	Hoists, Wire Rope, Electric Powered
MIL-T-21868	Truck, Lift, Fork, Diesel; General Specifications for
MIL-T-21870	Truck, Lift, Fork, Gasoline; General Specifications for
MIL-C-22137	Cranes Overhead Traveling, Overrunning, Electric Powered
MIL-T-22292	Truck, Lift, Fork, Weapons Handling, Electric Powered

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MIL-C-25200 General Requirements for Cable Assemblies, Special Weapons, Electrical

MIL-T-25959 Tiedown, Cargo Aircraft

MIL-H-26935 Hoists, Chain and Wire Rope, Electric Motor Driven

MIL-C-28546 Cranes, Overhead Traveling, Underhung, Electric Powered

MIL-S-45152 Semi-Trailer, Lowbed

MIL-C-38999 Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environment Resistant, Removable Crimp Contacts

MIL-T-45333 Trailers, Flatbed, 10 Ton, 4 Wheel, M345

MIL-T-45338, -45339, -45340, -45341, -45350, -45354 Trucks, Commercial

MIL-T-52740 Trucks, Lift, Fork, Gasoline, General Specifications for

MIL-A-85910 Airborne Stores and Associated Suspension Equipment; General Design Criteria for

DOD/DOE

AIRCRAFT MONITOR AND CONTROL (AMAC) SPECIFICATIONS SANDIA NATIONAL LABORATORIES DRAWINGS, KIRTLAND AIR FORCE BASE, ALBUQUERQUE, NEW MEXICO

a. Drawing No. 173838 Basic Design Specification, Fighter System A

b. Drawing No. 185435 PAL Fighter A

c. Drawing No. 185475 Basic Interface Requirement System 1

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STANDARDS

MILITARY

MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-461	EMI Characteristics; Requirements for Equipment
MIL-STD-462	Measurement of EMI Characteristics
MIL-STD-463	Definitions and System of Units; EMI Technology
MIL-STD-648	Design Criteria for Specialized Shipping Containers
MIL-STD-704	Electrical Power, Aircraft; Characteristics of
MIL-STD-882	System Safety Program for System and Associated Subsystem and Equipmnt; Requirements for
MIL-STD-1365	General Design Criteria for Handling Equipment Associated with Weapons and Weapon Systems
MIL-STD-1366	Definition of Packaging, Handling, Storage and Transportation, Systems Dimensional Constraints
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields; General Requirements for
MIL-STD-1399 (Navy)	Section 401, Interface Standard for Shipboard Systems, Section 401, D.C. Magnetic Field Environment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment for Facilities

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MIL-STD-1512	Electroexplosive Subsystems, Electrically Initiated, Design Requirements and Test Methods
DOD-STD-1679	Software Development
DOD-7935.1-S	Automated Data Systems Documentation Standards

HANDBOOKS

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures
MIL-HDBK-235	EM (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems

ORDNANCE DIRECTIVE

OD 44942	Weapon System Safety Guidelines Handbook

SPECIAL WEAPONS ORDNANCE PUBLICATIONS

SWOP 20-11	General Firefighting Guidance
SWOP 45-51	Transportation of Nuclear Weapons Material

TECHNICAL PUBLICATIONS

a. Wunsch, D. C., and Bell, R. R. "Determination of Threshold Failure Levels of Semiconductor Diodes and Transistors Due to Pulses Voltages," IEEE Transactions on Nuclear Science, Volume NS-15, No. 6, December 1968, pages 244-259

b. SC-DR-72 0492 "Approaches for Achieving Nuclear Weapon Electrical System Safety in Abnormal Environment," Sandia National Laboratories, Kirtland Air Force Base, Albuquerque, New Mexico 87115

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c. Tausworthe, Robert C., "Standardized Development Computer Software Part I - Methods, Jet Propulsion Laboratory," Pasadena, California, NASA Contract No. NAS 7-100, July 1976

d. Tausworthe, Robert C., "Standardized Development Computer Software Part II - Standards, Jet Propulsion Laboratory," Pasadena, California, NASA Contract No. NAS 7-100, August 1978

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SECTION 3
DEFINITIONS

3.1 ABNORMAL ENVIRONMENTS. Abnormal environments are those environments as defined in the weapon's Stockpile-to-Target Sequence and Military Characteristics in which the weapon is not expected to retain full operational reliability.

3.2 ACTIVATE SIGNALS. Those signals developed by AMAC/REL logic and delivered to the switching assembly where they are combined with 28-Vdc power to produce control signals.

3.3 AMAC/REL LOGIC. That logic within the multiplexed monitor, control, and release system which accepts command and monitor signals, performs Boolean operations on those signals, and outputs activate signals.

3.4 ARMED. The configuration of a nuclear weapon in which a single signal will initiate the action required for obtaining a nuclear detonation.

3.5 ARMED COMPONENT. A weapon component is armed when it has responded to an influence to change its state to enable a nuclear detonation.

3.6 ARMED WEAPON. A weapon is electrically armed when only a firing signal is required to discharge electrical energy into the warhead detonators.

3.7 ARM/DISARM (A/D) DEVICE. A mechanical or electromechanical device that provides a positive interruption of the firing circuit to prevent initiation of an explosive or pyrotechnic train prior to its commanded closures.

3.8 ARMING AND FUZING (A&F) SYSTEM. The aggregate of components within a nuclear weapon which generates signals to arm and detonate the warhead.

3.9 ARMING SYSTEM. That portion of a weapon which serves to ready (ARM), SAFE, or RESAFE (DISARM) the firing system and fuzing system and which may actuate devices in the nuclear system.

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3.10 AUTOMATA. That class of sequential machines which, by alterations of internal state, are capable of performing logical computational, or repetitive routines, such as automatic processors, computers, decoders, controllers, and where specifically denoted, their associated peripheral equipment.

3.11 CERTIFIED EQUIPMENT. Items of support equipment, combat delivery vehicles, and noncombat delivery vehicles which have received a nuclear safety engineering evaluation and have been nuclear safety approved for use with nuclear weapons and identified in appropriate publications approved by an agency of the US Navy.

3.12 CLASSIFIED CONTENT. Information which makes memory a critical component.

3.13 COMBAT DELIVERY VEHICLE. Any nuclear-capable vehicle and its installed equipment used for the combat delivery of nuclear weapons and whose operations are the subject of specified nuclear safety rules approved and promulgated by the Chief of Naval Operations.

3.14 CONTROL CURRENT. Electrical current in a nuclear weapon used to change the functional status of a component associated with prearming, arming, or detonating the warhead.

3.15 CONTROL SIGNALS. Those signals delivered by Multiplexed Monitor, Control, and Release Systems (MMCRS) to the Nuclear Weapon Interface (NWI) which result in prearming, safing, or suspension system unlock or release of a nuclear weapon.

3.16 COUPLING. A measure of data connectivity between a module and the program in which it is imbedded.

a. Common Coupling. Modules are common coupled if they reference data held in a common pool or central repository for certain data structures accessible by all the modules.

b. Control Coupling. Modules are control coupled if one module passes a flag or set of flags (control data) as arguments to the other, to directly influence the functioning of the receiving module.

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c. Data Coupling. Modules are data coupled if one module calls the other, if all input and output communication is in the form of arguments or parameters passed through the call-sequence interface, and if all such parameters are data (not control) elements.

3.17 CRITICAL. The use in this Handbook describes functions, circuits, activities, and hardware and software components which apply directly to, or control, the prearm, arm, fire, unlock, release, launch, or targeting functions of a nuclear weapon system.

3.18 CRITICAL COMPONENT. A component which requires application of the US Navy Two-Man Concept during an identifiable portion or all of its life cycle.

a. A nuclear weapon, with or without its fuzing and firing system installed.

b. A combat delivery vehicle which has had all preload functions completed and is ready for nuclear weapon mate or load.

c. A combat delivery vehicle which has a nuclear weapon(s) mated or loaded.

d. Hardware or software components designated by the Nuclear Weapon System Safety Group (NWSSG).

3.19 ELECTRICAL ISOLATION. Separation of electrical circuits, signals, or data to preclude ambiguity, interference, or information perversion. This may be achieved through physical isolation or by any property which distinguishes one electrical signal from all others (for example; time, phase, amplitude, or frequency).

3.20 ELECTROEXPLOSIVE DEVICE (EED). An electrically initiated explosive device within an electroexplosive subsystem having an explosive or pyrotechnic output, and actuated by an electroexplosive initiator.

3.21 ELECTROMAGNETIC COMPATIBILITY (EMC). The ability of electronic equipment, subsystems and systems to operate in their intended operational environments without suffering or causing unacceptable degradation because of unintentional electromagnetic radiation or response.

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3.22 ELECTROMAGNETIC INTERFERENCE (EMI). An electromagnetic disturbance which interrupts, obstructs, or otherwise degrades or limits the effective performance of electronics/electrical equipment. It can be induced intentionally, as in some forms of electronic warfare, or unintentionally, as a result of spurious emission and responses, intermodulation products, and the like.

3.23 ELECTROMAGNETIC PULSE (EMP). The electromagnetic radiation from a nuclear explosion caused by Compton-recoil electrons and photoelectrons from photons scattered in the materials of the nuclear device or in a surrounding medium. The resulting electric and magnetic fields may couple with military systems to produce damaging current and voltage surges.

3.24 ELECTROMAGNETIC RADIATION (EMR). Radiation made up of oscilating electric and magnetic fields and propagated with the speed of light. Includes gamma radiation, X-rays, ultraviolet, visible and infrared radiation, and radar and radio waves.

3.25 FAIL-SAFE. A design feature of a nuclear weapon system/component which ensures that, under failure, no critical function, damage to equipment, or injury to personnel will occur.

3.26 FIRMWARE. Logic circuits in read-only memory or specifically protected portions of memory that may be altered by the software only under certain circumstances.

3.27 HARD-WIRE. A dedicated discrete electrical circuit.

3.28 LIMITED CURRENT. Monitor or test currents which are limited so that the most current that can be delivered to a nuclear weapon for monitoring or testing purposes will be less than that required to activate the most sensitive component in the arming sequence.

3.29 LOGIC CIRCUITS. Interconnected electrical, electronic and microelectronic circuits that carry and control electrical impulses, for the purpose of providing the "ON-OFF" intelligence for predetermined relationships within a system or subsystem. In many cases, these circuits perform functions which follow the laws of Boolean algebra.

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3.30 LOGISTIC MOVEMENT. The transport of nuclear weapons in connection with supply or maintenance operations. Under certain specified conditions, combat aircraft may be used for such movements.

3.31 LOGISTIC SYSTEM. The organization, noncombat delivery vehicles, and support equipment employed for loading, movement and transfer of nuclear cargo.

3.32 MODULE. An identifiable, self-contained, compilable, independent set of computer instructions with one entry point and one exit point that fulfills some well-defined function. A module is a combination of one or more units.

3.33 MONITOR CURRENT. A limited current introduced into a nuclear weapon to determine the functional status of selected components.

3.34 MULTIPLEXED SYSTEM. A signal transmission system in which two or more signals share one transmission path or bus. A multiplexer is that portion of the miltiplex system which formats, transfers, and processes signals required for nuclear weapon monitoring, control, and separation.

3.35 NONCOMBAT DELIVERY VEHICLE. Any vehicle and its installed equipment used for the movement, loading, or transfer of nuclear weapons, but not the subject of specific nuclear safety rules.

3.36 NORMAL ENVIRONMENT. The expected logistical and operational environments, as defined in the weapon's stockpile-to-target sequence in which the weapon is required to survive without degradation in operational reliability.

3.37 NUCLEAR CARGO. A nuclear weapon or nuclear component (except Limited Life Components (LLC)) prepared for logistic movement.

3.38 NUCLEAR CONSENT SIGNAL. An electrical enabling signal(s) originated by a conscious, deliberate human act signifying intent of release or launch of an unarmed or prearmed weapon.

3.39 NUCLEAR SAFETY CRITERIA. Design and evaluation criteria established as aids for ensuring that nuclear safety is a basic system engineering and procedural requirement in nuclear weapon systems and logistics systems.

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3.40 NUCLEAR SAFING. The influence of a nuclear warhead design feature which prevents a significant nuclear yield as a result of detonating the high explosive (HE) by any means other than the intended firing system.

3.41 NUCLEAR WARHEAD. An item which normally consists of the explosive system, the nuclear system, and electrical circuitry. It may be of an implosion type, gun type, or thermonuclear. Nuclear components may be integrally contained. It is usually designed to be mounted in or on a torpedo, guided missile, rocket, bomb, or similar delivery system.

3.42 NUCLEAR WEAPON.

a. A device in which the explosion results from the energy released by reactions involving atomic nuclei, either fission or fusion, or both.

b. A nuclear bomb, warhead, reentry system, reentry vehicle, or payload section containing a nuclear warhead (includes guided air-launched missiles).

3.43 NUCLEAR WEAPON INTERFACE (NWI). A system interface through which information is passed between MMCRS and a nuclear weapon and the in-flight operable bomb rack lock (IFOBRL)/weapon release mechanism.

3.44 NUCLEAR WEAPON SYSTEM. A combat delivery vehicle with its nuclear weapon(s) and associated support equipment, noncombat delivery vehicles, facilities, and services.

3.45 ONE-POINT SAFE. A term used to describe the degree of safety in a nuclear weapon. A characteristic of a nuclear weapon which, upon undergoing one-point detonation initiated anywhere in the high explosive (HE) system, has a probability of no greater than one in one million of producing a nuclear yield in excess of 4 pounds of Trinitrotoluene (TNT) equivalent.

3.46 PHYSICAL ISOLATION. Physical separation of wiring, parts, modules, assemblies, and so forth, to assure no physical contact or interaction.

3.47 PREARMED. The state of a weapon system in which a single launch, release, or command signal will irreversibly initiate the arming/firing sequence.

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3.48 PREARMING. A function(s) initiated by an operator(s) which places the weapon in a prearmed state.

3.49 SAFE/ARM (S/A) DEVICE. A device which provides electrical interruption of the firing circuits and mechanical interruption between the initiator and the subsequent explosive or pyrotechnic train.

3.50 SIGNIFICANT NUCLEAR YIELD. The energy released through nuclear fission or fusion equivalent to or greater than the energy release by detonation of 4 pounds of TNT.

3.51 SOFTWARE. Information used to control or program, and which is processed or produced by automatic machines. It may be stored or transferred on a variety of informational storage media. Software, as used in the context of this instruction, includes firmware.

3.52 STOCKPILE-TO-TARGET SEQUENCE.

a. The order of events involved in removing a nuclear weapon from storage, and assembling, testing, transporting, and delivering it on the target.

b. A document which defines the logistical and employment concepts and related physical environments involved in the delivery of a nuclear weapon from the stockpile to the target.

3.53 STRAY VOLTAGE. An extraneous voltage on a deenergized circuit.

3.54 STRUCTURED DESIGN. A software hierarchial design technique with specified rules governing the hierarchy in terms of design order of modules, independence of modules, and specification of data and interfaces.

3.55 STRUCTURED PROGRAM. A program whose control logic topology adheres to strict rules of form, being composed of interactions and nestings of a set of basic planar flowchart constructions: IFTHENELSE, WHILEDO, DOWHILE, CASE, and sequence formats, along with, perhaps, paranormal extensions.

3.56 SUPPORT EQUIPMENT. The equipment ancillary to a nuclear weapon for the purpose of handling and maintaining, as well as

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that associated test equipment required to verify the functioning of the nuclear system. (General and special purpose hand tools such as pliers, wrenches, screwdrivers, and so forth, are specifically excluded.)

3.57 UNAUTHORIZED LAUNCH. Any missile launch or bomb release resulting from unauthorized acts. The missile/bomb need not follow a controlled trajectory, nor reach any specified impact point.

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SECTION 4
GENERAL

4.1 DESIGN AND EVALUATION CRITERIA:

4.1.1 Minimum Standards for Design. The design criteria outlined here are the minimum nuclear safety standards for design of nuclear weapon systems. These criteria are not considered design solutions nor all-inclusive nuclear safety standards and are not intended to restrict the designer in the methods and techniques used to meet operational design requirements. Because these criteria are not all-inclusive, the designer may still be obligated to introduce additional safety features as the individual situation dictates.

4.1.2 Minimum Standards for Certification. The evaluation criteria outlined here establish the minimum safety standards for certification, provided no other features are introduced tnat would negate the total design safety requirements. These criteria do not invalidate other safety requirements of the Department of Defense (DOD) publications. However, Navy activities and contractors will apply the more stringent criteria given here.

4.1.3 Minimum Standards for Specific Items. Requirements and criteria for design and evaluation of specific items are based on the uses specified in formal documents such as the Operational Requirements (ORs), Operational Concept, Program Management Directive (PMD), Stockpile-to-Target Sequence (STS), and the Military Characteristics (MCs) for nuclear weapon/warhead or nuclear weapon system development.

4.2 NUCLEAR WEAPON SYSTEM SAFETY. Because of their political and military importance, destructive power, cost, and the consequences of an unauthorized or accidental nuclear or high explosive detonation, nuclear weapons must be protected against the risks and threats inherent in their environment.

a. The conservation of nuclear weapons as a national resource and the safety of the public, operating personnel, and property are of paramount importance. These weapons and weapon systems must be designed to incorporate maximum safety consistent with operational requirements.

b. Nuclear safety requirements are implemented to prevent nuclear accidents and to minimize both the number and consequence of nuclear incidents and deficiencies.

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4.3 DOD NUCLEAR WEAPON SYSTEM SAFETY STANDARDS. The DOD has established safety standards which are the basis for the nuclear safety rules governing nuclear weapon systems operations. The system designer should bear in mind that the nuclear safety standards must be met, and that the operational use of a total nuclear weapon system may be restricted if design safety criteria are not satisified. These standards require that, as a minimum, there shall be positive measures to:

a. Prevent nuclear weapons involved in accidents or incidents or jettisoned weapons from producing a nuclear yield.

b. Prevent deliberate prearming, arming, launching, firing, or releasing of nuclear weapons except upon execution of emergency war orders or when directed by competent authority.

c. Prevent inadvertent prearming, arming, launching, firing, or releasing of nuclear weapons in all normal and credible abnormal environments.

d. Ensure adequate security of nuclear weapons pursuant to the provisions of DOD Directive 5210.41.

4.4 SAFETY DESIGN CRITERIA. The following safety design criteria apply:

a. The DOD nuclear weapon system safety standards must be met.

b. The malfunction or accidental operation of any single software or hardware component must not result in the prearming, arming, launching, firing, or releasing of a nuclear weapon.

c. Designers must emphasize human engineering to minimize the probabilities of human error.

d. Adequate nuclear safety must be designed into the system so that dependence on administrative procedures for providing safety can be minimized.

e. The system must be designed to minimize the probability that a single individual, with access, can prearm, arm, launch, fire, or release a nuclear weapon.

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SECTION 5
COMBAT DELIVERY AIRCRAFT SYSTEMS

5.1 GENERAL. The design criteria outlined here apply to aircraft delivery, launch, suspension, release, and weapon monitor weapon control systems. (The safety devices required by these criteria may also be used for nonnuclear stores except where specifically restricted from such use.) Command and control devices, if used, will be in addition to these requirements. Integrated armament stores management systems that meet these criteria may be used. Designs for multiplexed stores management systems must meet the requirements for safety outlined here. The aircraft nuclear weapon system must be designed to include the following:

a. No single software or hardware component failure or system fault, or normal environmental condition will allow any of the following functions to occur:

(1) Operation of the in-flight operable bomb rack lock (IFOBRL) (paragraph 5.2.g.).

(2) Circumvention (deliberate or accidental) of two-man control for prearming, release, or launch (paragraphs 5.2.f. and 5.3.a.).

(3) Availability of unintended power at the aircraft/nuclear weapon electrical interface.

(4) Accidental separation of nuclear weapon from the aircraft.

b. No critical function in a. above can occur by the opening or closing of any circuit breaker or other circuit protective device. Operating power or control functions which can directly cause operating power to excite the critical functions shall not be connected to a device (such as semiconductors) whose failure mode could cause operating power to be applied.

c. Nuclear weapon monitor, control, and release functions shall have:

(1) A control or control setting dedicated to prearming the nuclear weapon(s) (paragraph 5.3.b.(9)).

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(2) A control or control setting dedicated to releasing or launching a nuclear weapon(s) (paragraph 5.2.e.).

(3) A control for unlocking the IFOBRL (paragraphs 5.2.e. and 5.2.g.).

(4) In multiplace aircraft, a two-man control for prearming and release or launching (paragraphs 5.2.f. and 5.3.a.).

(5) In any multiplexed stores management system hard-wire discrete consent functions are required for weapon prearming and unlocking the IFOBRL to prevent these functions from occurring even though the command for such functions has been transmitted over the multiplex bus. Circuits for these consent functions shall be physically and electrically isolated from all other circuits and from each other. An exception to these hard-wire consent function requirements may be granted where adequate statistical proof is offered to assure that the likelihood of inadvertently prearming the weapon or unlocking the IFOBRL is less than 10^{10} during the lifetime of the weapon system or is less than 1 in 10^7 for any occurrence of a catastrophic event/abnormal environment.

d. Monitor and control functions, unlocking devices, release devices, and Safe/Arm (S/A) and Arm/Disarm (A/D) devices shall be powered from electrical bus/buses which are powered from secondary or backup power source(s) if the primary power source is lost.

e. The probability of activating critical functions during catastrophic events/abnormal environments shall be minimized. Catastrophic events/abnormal environments include air or ground collision, crash, fire, explosions, excessive current flow, and severe environmental conditions. In no case shall a system likelihood of nuclear detonation greater than 1 in 10^6 be permitted, given the occurrence of such a catastrophic event/abnormal environment.

f. Aircraft electrical power failure shall not jeopardize the safe condition of a weapon.

g. Jettison or release of a safe weapon with all pertinent nuclear weapon circuits and components in the deenergized or deactivated condition must be possible.

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h. Electrical power is applied on a designated pin(s) of the weapon interface connector at electrical separation of the weapon from the aircraft for an armed release. These pins are identified in the Aircraft Monitor and Control (AMAC) Specifications for bombs. Missiles are exempt from this requirement.

i. In no case shall the system likelihood of inadvertent nuclear detonation be greater than 1 in 10^9 for the lifetime of each nuclear weapon for the normal stockpile-to-target sequence environments.

5.2 NUCLEAR WEAPONS SUSPENSION AND RELEASE SYSTEMS. These systems must be designed so that:

a. The suspension system shall prevent weapon separation, release, ejection, launch, or jettison by any other means except the proper operation of control devices. Mechanical cables, if used, must be protected from accidental operation.

b. Suspension and release items (rack, rail, and so forth) shall not have electrical power applied which could cause the operation or firing of any device, until the start of release preparations.

c. Circuits used to cause release of nuclear weapons shall be electrically isolated from all other functions.

d. The latched and locked condition of the suspension devices shall be positively determined visually and electrically while the aircraft is on the ground.

e. Release systems must have at least one separate control or control setting peculiar to the release or launch of nuclear weapons (in addition to the control for the IFOBRL); this control or setting will not be used to release or launch non-nuclear weapons, other than nuclear weapon training items.

f. In aircraft requiring a crew of more than one person, the release actuating controls and the IFOBRL controls must be separated so that the release or launch of a nuclear weapon requires the action of at least two crew members. For multiplace aircraft used in combat by one person, the system may be designed to permit single position operation only after the second crew position function has been accomplished prior to flight. However, this function must not bypass the IFOBRL.

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g. An IFOBRL is provided which, when locked, will prevent weapon release even though the normal releasing force is generated and transmitted to the release system. The lock and its controlling mechanism must be part of the nuclear weapon release system. The IFOBRL system shall:

(1) Mechanically constrain the releasing device.

(2) Prevent release or launch in the event that maximum available release force is accidentally applied in the release mechanism.

(3) Fail-safe, if failure occurs when the lock is locked.

(4) Disable all means of release when in the locked position.

(5) Permit a visual check of the locked state by ground personnel. For direct visual inspection, the locking device itself must present an unmistakable indication of the locked state.

(6) Be protected from accidental operation.

(7) Provide a method in the crew compartment to reveal tampering with the cockpit control of the IFOBRL.

(8) Provide a remote indication to the aircrew that indicates the fully locked and/or unlocked positions of the IFOBRL. If a single indication for the locked condition is used, it must reflect only the fully locked position of the IFOBRL. If a single indication for the unlocked condition is used; it must reflect every condition other than a fully locked condition. A failure in the remote indication system must never allow an apparent indication to the aircrew of a locked condition when, in fact, an unlocked condition exists.

(9) Provide a lock indication system which is independent of electrical connections between the aircraft and weapon.

(10) If hooks are used in the suspension and release linkage, the safety lock shall mechanically restrain the suspension and release linkage. If hooks are used that can be individually latched or unlatched, the safety lock must provide mechanical restraint to each hook.

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h. Pylons carrying nuclear weapons must meet one of the following criteria:

(1) Pylons shall not be jettisonable; or

(2) Pylon jettison system shall include a lock which meets the criteria for the nuclear weapons lock stated in paragraph 5.2.g. above. A single lock may be used for both the weapon and the pylon where such a design is feasible.

i. If a jettison mode is provided, the suspension and release system shall allow jettison of weapons in a safe or prearmed state with prearming dependent upon all normal prearming criteria being fulfilled.

j. The discrete signals for unlocking the IFOBRL for release must remain physically and electrically isolated from each other, coinciding only at the release device. No release system malfunction shall be capable of operating the IFOBRL nor shall a malfunction in the IFOBRL cause a release. The operation of the IFOBRL shall not cause a release of a weapon from a release device which has received a release signal prior to the unlock signal (i.e., it shall permit a release only when a release signal follows the unlock signal).

5.3 NUCLEAR WEAPONS MONITOR AND CONTROL. The monitor and control system must be designed to prearm and safe nuclear weapons and monitor the weapon. The following safety criteria apply:

a. In combat aircraft requiring a crew of more than one person, prearming controls must be separated so that prearming the weapon requires independent actions by at least two crew members. For a multiplace aircraft which may be used in combat by one person, the system may be designed so that the weapon can be prearmed from a single position only after the second crew position function has been accomplished before flight.

b. The weapon monitor and control system must be designed so that:

(1) Prearming functions are reversible.

(2) An aircraft electrical power failure shall not change the safe condition of a weapon.

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(3) Prearming control circuits and safing circuits are electrically isolated from all other circuits.

(4) Prearming power circuits are physically and electrically isolated from all other circuits, however, the power and control circuits may coincide at the final switching assembly prior to the nuclear weapon interface. Physical isolation should be maintained to the maximum extent possible within the final switching assembly.

(5) Availability of electrical power inside the switching assembly shall be dependent upon a control function and/or command peculiar to the nuclear weapon control and monitor system.

(6) Power sufficient to operate any critical weapon component cannot be applied until activation of that component is required. Each monitor circuit current shall be limited so as not to exceed milliamperes. Currents in excess of 100 milliamperes will not be applied to weapons until control power is deliberately applied.

(7) Monitor circuits must be electrically isolated from power and control circuits. Monitor functions shall be capable of being performed independent of weapon control functions.

(8) It must be possible to determine the safe condition of the weapon system through continuous or on-demand monitoring.

(9) Application of prearming or safing control power to a weapon shall require a control or control setting peculiar to the nuclear weapon(s) that requires a separate, deliberate, and conscious act by the weapon system operator.

c. When electrical prearming power is applied, an explicit indication of the prearmed or safe condition of the weapon(s) must be provided continuously.

d. Nuclear prearming controls shall be designed to provide for a method of detection of tampering.

5.4 NUCLEAR CAUTION SYSTEM. Nuclear caution system monitoring requirements for the suspension and release system and the nuclear weapon monitor and control system must bring the following occurrences to the attention of the flight crew:

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a. Unlocking of or an unlock signal to the IFOBRL when unlocking has not been commanded by normal operation of controls.

b. Prearming of a weapon when the weapon prearming has not been commanded by normal operation of controls.

c. The safe status of the weapon cannot be positively determined. A delay may be designed into this warning function so that a caution will not be received by the flight crew during weapon change of state from safe to prearm (or vice versa).

d. Nuclear release signals not commanded by normal operation of controls.

5.5 AIR-LAUNCHED MISSILES AND ROCKETS. An air-launched nuclear missile or rocket is generally considered as an extension of the aircraft until it is launched or released. Therefore, the same criteria that apply to aircraft (such as connector design, EMR protection, electrical subsystems, and so forth), must be applied to the missile or rocket. Additionally, the following criteria shall apply:

a. A rocket motor must have an S/A device which, in its safe position, protects against accidental firing. The device shall conform to the standard definition for an S/A device. For other missile propulsion systems, an A/D device is acceptable in place of the S/A device.

b. The safe and armed position of each S/A or A/D device will indicate the safe or armed condition to the aircrew. A single indication of ARMED may be used to reflect any condition other than an entirely safe condition.

c. Mechanical locking devices used with the S/A and A/D devices shall be located so that they may be removed at the last practical point in either the missile assembly or missile loading sequence.

d. The suspension and release system must provide for an IFOBRL. For rail-launched weapon where it may not be practicable to use an independent lock, compensatory safety devices must be used. These devices must offer the same protection against accidental launch or release that the IFOBRL offers to freefall or ejected weapons, and must include:

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(1) A launch circuit interruption device located in the aircraft as close as possible to the entry point of the launch circuit into the missile or rocket.

(2) A control in the crew compartment which provides for interruption of the launch circuit. A method to reveal tampering with the control shall be included.

e. The signal to launch a missile or rocket shall be generated separately and electrically isolated from the signal that activates the S/A or A/D device. The two signals shall coincide only at the missile/rocket.

f. No other system, in its operational or failure mode, will be capable of actuating the missile or rocket propulsion system.

g. Rocket motor firing circuit connectors will not contain spare or unused pins.

5.6 AUTOMATA.

a. Combat aircraft systems automata, in addition to paragraphs 5.1., shall be designed to ensure that:

(1) Operators can detect abnormalities and take proper and timely corrective action to control critical functions.

(2) The automata will not automatically bypass the operator commanded controls required in this section (nuclear consent, prearming and release/launch controls, and IFOBRL).

b. If software is used to control critical functions, a software nuclear safety analysis shall be required (see paragraph 6.4).

5.7 AIRCRAFT LOADING AND UNLOADING PROCEDURES. See paragraph 8.2.4 for criteria governing loading and unloading procedures.

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SECTION 6
NUCLEAR WEAPON SYSTEM AUTOMATA

6.1 GENERAL. The design criteria outlined in this section apply to automata which receive, store, process or transmit data to prearm, arm, enable, unlock, fire, launch, or release a nuclear weapon. Such equipment shall be designed to provide the highest degree of protection against accidental or unauthorized activation of critical command and control functions.

6.2 AUTOMATA DESIGN CRITERIA

6.2.1 Software Design and Development. All system software, including any support software and test maintenance software, shall be designed and coded using structured design techniques and structured programming techniques, and shall be in complete conformance to MIL-STD-1679.

6.2.2 Module Coupling. The software shall be broken down into independent compilable modules which are completely uncoupled or if coupling is necessary, only the lowest levels of coupling shall be used, i.e., data coupling, and/or common coupling.

6.2.3 Documentation. The Software Requirements, Design, and Implementation shall be documented as early as possible and shall comply with the documentation requirements of DOD-STD-1679 (Navy) and Department of Defense Automated Data Systems Documentation Standards 7935.1-S. All documentation shall be updated to reflect changes and progress throughout the life cycle.

6.2.4 Critical Function Commands. Any critical function commands issued or controlled by software such as prearming, arming, releasing, firing, and safing commands shall be designed to preclude inadvertent and accidental generation of these commands from complementary bit patterns or from unexpected or uncontrollable bit errors in the command word or data stream. Techniques such as Longitudinal Redundancy Checks (LRC) and Cyclic Redundancy Checks (CRD) may be used for disk type peripherals. Double error checks and correction techniques for memory devices and non-linear check summing techniques shall be used where appropriate.

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6.2.5 Program Loading. The system shall be designed to prevent automatic control until all valid and correct program data have been loaded and verified. If the operator controls the loading, a method will be provided to halt the loading if any error occurs and prevents use of the data if proper loading is not verified.

6.2.6 Memory Data Change. Once the memory has been loaded and verified, the data shall be protected from unauthorized or accidental change. Memory change due to hardware failure shall be minimized to the maximum extent possible. A single hardware fault must not result in a memory change involving critical functions. Reloading or changing the memory critical contents shall require a method for proper entry to the memory. Any attempt to gain access to memory to load or change it without the proper discriminatory means for entry shall cause a positive indication to the weapon system operator or maintenance crew and the automata entry will be automatically rejected. If selected computer software routines are stored at manned control centers for future transfer to unmanned weapon system facilities by the communication system, a method must be devised to ensure that routines have been transferred correctly. A method must also be devised to prevent unauthorized or inadvertent data transfers by anyone who has access to the communication system.

6.2.7 Memory Security. The system shall be designed to preclude unwarranted/unauthorized recovery of classified information from its memory. System elements shall be classified at a level commensurate with the highest level of classified information content. A method will be provided to permit downgrading of the classified contents of the memory pertaining to nuclear safety. The method must preclude meaningful data retrieval without the use of specialized, nonportable laboratory equipment. System operating instructions must specify that if this security measure is ineffective, memory units will be classified at the highest category of the information processed.

6.2.8 Control. Automata which contain nuclear weapon system critical logic will be designed so that a deviation from proper routine or sequence will automatically result in a recycle to the proper sequence or a self-test mode. If this requirement cannot be met, an automatic shutdown with an associated signal to the operator will be provided.

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a. If automata are entirely program controlled, safety features shall be incorporated to prevent improper sequences of routines from occurring.

b. Detection of improper processing shall be provided in all cases where continuaton of automatic program control would degrade nuclear safety.

6.3 HARDWARE CROSS-CHECK. The design of automata for critical functions shall provide self-check, confidence, or test routines. Such automata shall have a first or second level interface with a nuclear weapon (paragraphs 6.4.2.a. and 6.4.2.b.) and thus, if susceptible to tampering, comprise an inherent unauthorized action fault. Automata design may incorporate one of two methods of cross-check:

6.3.1 Special Test Circuitry. This circuitry is designed to self-test the hardware and shall disclose tampering.

6.3.2 Certification/Confidence Programs. Software programs may be specially developed to check critical circuits. Software checks must be designed so that routines within the programs will indicate the possibility of unauthorized actions.

6.4 SOFTWARE NUCLEAR SAFETY ANALYSIS (SNSA). Any software (or methods of weapon control) used to process the data for, provide the status of, or which can exercise automated/automatic control over any critical nuclear weapon system function shall be subjected to an SNSA to ensure system nuclear safety integrity.

6.4.1 Purpose and Requirements. The SNSA provides assurance that the software or other program controls as designed, coded, and implemented, cannot contribute to accidental, unauthorized, or fault activation of critical nuclear weapon system functions.

6.4.2 Guidance. Software components will be designated for a SNSA based upon an analysis of the degree to which the software could affect critical functions, or contribute to an authorized launch or release. Software is categorized by degree of involvement as follows:

a. First Level Interface. Any software used by automata having a direct electrical connection to a nuclear weapon is categorized as a first level interface. This includes all resident or processor-accessible programs, even though no critical functions are performed.

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b. Second Level Interface. Any software used by automata having a direct electrical connection to automata having a first level interface is categorized as a second level interface. This includes all resident or processor accessible programs even though no critical functions are performed.

6.4.3 Software Currency Requirements. Nuclear safety critical software analyses shall be conducted throughout the weapon system's life cycle whenever new or modified software is developed. Modified software may have a modified SNSA covering the changed segments, but all interfaces to the original program must be checked. Software which is not subject to recurring changes requires only an initial review, analysis, and certification.

6.4.4 Special Diagnostic Software. Certain software for maintenance or diagnostic purposes may require an SNSA.

a. Diagnostic software planned for use must receive an SNSA if the appropriate computer memory overwrite is not planned to be loaded before the diagnostic tape.

b. Diagnostic software planned for use must receive an SNSA if cryptographic codes, variables, and routines in the computer memory are not planned to be overwritten or erased before the diagnostic software is loaded.

c. Diagnostic software which have received an SNSA shall be maintained under continuous two-man control.

d. Positive measures must be taken to preclude the possibility of residual diagnostic software instructions remaining in memory after completion of any maintenance or diagnostic tests.

6.4.5 SNSA Agencies. The organization performing the SNSA and the software developer organization must be managerially and financially independent of each other. In addition, the two organizations must not be in direct contact except under the specific direction and control of the Program Manager. If the software developer is a US Government agency, the SNSA may be performed by a functionally distinct department within that agency. The organization or contractor that developed the software being analyzed must not develop any software tools employed in the SNSA. The Program Manager must qualify such

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tools to determine their suitability for use in the SNSA Program. The integrity of software analysis tools must be assured through proper security and control procedures. The SNSA shall be conducted and documented to verify that:

a. The software to be checked is identical to the engineering original or master.

b. Critical areas, routines, program branches, switches and commands within the software are identified and checked.

c. No unauthorized programming is present that could affect performance of nuclear safety critical functions.

d. Under all specified input conditions, the program response is completely consistent with nuclear safety design and evaluation criteria.

e. A high level of confidence is achieved under all operational and maintenance conditions that the program will not cause or contribute to the improper performance of the critical functions which affect the safety of the nuclear weapon system.

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SECTION 7
ARMING AND FUZING (A&F) SYSTEMS

7.1 ARMING AND FUZING DESIGN CRITERIA. An arming and fuzing system is the aggregate of components, devices, and design features which generates signals for weapon arming, fuzing, and detonation, as well as those components and features which protect against unauthorized or accidental arming, fuzing, and detonation. The guidance in this section is intended for the use of Navy and Navy contract designers and evaluators. Both DOD and Department of Energy (DOE) subsystems are normally a part of the total weapon system arming and fuzing design.

7.1.1 Weak Link/Strong Link Design. Nuclear safety devices and features must be designed to withstand abnormal environments for a longer time period than the critical arming components or until the warhead is physically rendered incapable of providing a nuclear yield. The following characteristics are integral parts of the DOE nuclear weapon system design concept. These concepts should be considered for inclusion in the design of Navy developed portions of arming and fuzing systems:

a. Exclusion Region. This special region contains the firing set and weapon detonator system and the necessary packaging and safety devices to exclude electrical energy from the firing set and weapon detonator system except in an intended use situation.

b. Weak Links. A weak link is a selected element vital to the operation of the firing set and weapon detonator system whose function is not likely to be duplicated or bypassed (for example; capacitor, transformer). Weak links are designed to respond predictably to certain levels and types of abnormal environments by becoming irreversibly inoperative thus rendering the system inoperable, at levels less than those at which the strong links fail to maintain electrical isolation.

c. Strong Links. Access to the firing set for the arming and firing signals is through a group of safety devices called strong links, which includes system enable (intent) switches and environmental sensing devices. These strong links are designed to provide electrical isolation in abnormal environments and to actuate in their normal mode only in a situation where their use is intended.

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7.1.2 Devices.

a. Environmental Sensing Device (ESD). The arming and fuzing system shall contain an ESD which will prevent a critical function until it is activated by the unique environment peculiar to intended release or flight. At least one ESD will provide strong link abnormal environment protection.

b. A/D or S/A Device. All nuclear weapons systems shall contain an A/D device (sometimes designated as the ready/safe switch) or an S/A device. It shall be designed so that it may be electrically armed and electrically and/or manually safe (it must not permit manual or accidental arming during safing). It must permit a check of the safe/arm status. This status may be electrically monitored during phases of system life when visual observation is not practical. The device will provide strong link protection.

c. Intent Switch. The arming and fuzing system shall contain an intent switch. The intent switch is a strong link safety device that is activated by a unique prearming signal. This signal should be derived from some portion of the weapon system under human control. The intent switch may also serve as the A/D or S/A device.

d. Launch/Release Sensing Device. For aircraft delivery systems, the arming and fuzing system shall contain a device (for example, a pullout switch or breakaway connector) which will isolate the arming and fuzing system electrically and mechanically from any arming power source until a mechanical force is applied to a switch during the release or launch operation.

7.1.3 Arming and Fuzing Features.

a. Abnormal Environment Protection. Abnormal environment protective features shall prevent arming in all abnormal environments specified in the applicable stockpile-to-target sequence.

b. Dual Signal Arming. At least two separate and independently derived signals (which cannot be generated by a single signal at any point) shall be required to arm the weapon. Both required signals must be interrupted by a strong link device(s).

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At least one of these signals shall be continuous after application. This is not a requirement for multiple power sources. The probability that the arming and fuzing system will provide these signals prematurely prior to prearming, shall be less than one event in one billion per individual weapon lifetime in a normal environment, nor one event in one million per occurrence in an abnormal environment. After the arming and fuzing system has been prearmed, the probability of premature arming due to arming and fuzing system failure must not exceed one event in ten thousand.

c. Automatic Disarm. The arming and fuzing design must provide for the automatic discharge of stored energy (excluding batteries) in the event arming power is interrupted.

d. Unique Signal Prearming. The arming and fuzing system shall require a unique prearming signal that will prevent accidental operation by stray voltage, short circuits, and so forth. Unique signals will actuate the intent switch or other specific safety-critical load in the weapon system. ("Unique" means that only one signal source in the weapon system can actuate the corresponding load from its normally safe condition.) The signal format is chosen to allow discrimination of the unique signal from any other signal in the weapon system. Its source and format are designed to preclude inadvertent or accidental generation in an abnormal environment.

e. Lightning Protection. Lightning protection and EMR filters must be incorporated so that critical weapon arming and fuzing circuits are protected throughout the stockpile-to-target sequence.

f. Grounding. Grounding practice shall not permit the bypass of safety features in any environment. Structure or case grounding will not be used for critical components unless the practice can be shown to enhance nuclear safety.

g. Cable Damage Protection. The design shall minimize the probability of cable damage during assembly, maintenance, and operation.

h. Nondestructive Testing Compatibility. The arming and fuzing system design shall be compatible with standard Navy nondestructive testing techniques which may be used on the weapon system.

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i. Chemical Compatibility and Reversion. All material used shall be chemically compatible within the stockpile and storage environment. No materials shall be used in the design which have the capability to increase the high explosive sensitivity, generate explosive gases, cause electrical shorts, revert, and so forth.

j. Isolated Circuits. Arming and fuzing circuits must be physically and electrically isolated from all other circuits unless incompatible signal formats are used.

k. Monitoring Requirements. The capability to monitor at least one critical safety component in the arming and fuzing system for safe condition either continuously or on demand is required.

l. Monitoring Current Compatibility. All prearming and arming components must not operate when exposed to system monitoring current. Monitor currents must not exceed 100 milliamperes under any conditions.

m. Explosive Ordnance Disposal (EOD). Arming and fuzing design shall facilitate emergency EOD operations.

n. Human Engineering Practice. The arming and fuzing design shall incorporate human engineering characteristics to minimize personnel error.

o. Input/Output Isolation. Isolate the electrical inputs to nuclear safety devices from the outputs, or use other acceptable techniques to minimize the effects of short circuits.

p. Electroexplosive Device (EED) Protection. These devices shall be protected from inadvertent or accidental signals.

7.2 S/A AND A/D DEVICES. See MIL-STD-1512.

7.3 MISSILES/ROCKETS. When a DOE warhead is adapted to a weapon system for which the Navy has the responsibility for design of the arming and fuzing system, the same criteria that apply for the arming and fuzing systems (such as ESD, A/D, launch/release sensing device, and arming and fuzing design features) also apply to the adaption kit or missile/rocket system which performs such functions with the exception of the following:

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7.3.1 Environment Sensing Device. There is no requirement for monitoring the warhead ESD at the missile/rocket or aircraft interface.

7.3.2 Missile SAF Arming/Safing Device. The missile/rocket SAF system device (such as the A/D or S/A device) used for arming/safing the missile/rocket SAF system shall require weapon system operator action for application of safing power.

7.3.3 Launch/Release Sensing Device. The launch/release sensing device shall isolate the missile/rocket arming and fuzing system electrically and mechanically from any arming power source until a mechanical force is applied to the device during launch operation.

7.3.4 Lightning Protection. Lightning protection must be incorporated for critical weapon arming and fuzing circuits.

7.3.5 Monitoring Requirements. The capability for monitoring the missile/rocket SAF arming/safing device (paragraph 7.3.2) for SAFE condition either continuously or on demand is required.

7.3.6 Connector Pin Assignments. The assignment of functions to pins within a connector will ensure that a short circuit occurring as a result of a bent or misaligned pin will not result in delivery of more than 100 milliamperes current to the warhead.

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SECTION 8

NONCOMBAT DELIVERY VEHICLES AND SUPPORT EQUIPMENT

8.1 GENERAL. Safety must be incorporated into the design of noncombat transport vehicles and support equipment used to move, store, load, and unload nuclear weapons. The safety design factors must allow for the uncertainties in predicting operational conditions, variations in material strengths and in manufacturing, and uncertainties introduced by simplified design procedures. These criteria supplement good industrial design practices, standards and features, and are not intended to prohibit the use of any commercial design off-the-shelf vehicle or equipment procured for nonnuclear purposes (such as trucks, truck tractors, semitrailers, trailers, cranes, hoists, winches, and so forth) which meet these criteria. The equipment must meet appropriate structural, environmental, and mobility requirements. The applicable STS will define modes of transportation. All equipment must be designed to operate under conditions described in the operational concept or logistic sequence for the equipment or the major system.

8.1.1 Ground Transportation Equipment. This equipment shall be designed so that the nuclear weapon is not supported by the lift arms, cables, or hydraulic systems, but is supported by the basic frame of the equipment. (This does not apply to equipment used solely for positioning or transferring nuclear weapons in a designated area.)

a. Static grounding provisions must be considered for equipment designed for specific nuclear weapon systems.

b. The equipment shall be designed so that it will reduce the transmission of fire (electrical or fuel system failure) or shock to the nuclear weapon(s).

c. Equipment that will carry nuclear weapons and will be transported by an auxiliary vehicle during air and ground operations shall be designed so that the nuclear weapon is not supported by the transport equipment lift arms, cable, hydraulic systems. The weapon shall be supported by the basic frame of the equipment or the auxiliary ground transport vehicle structure.

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8.1.2 Structural Design. This structural design applies to weapon loaders, forklift, trailers, and self-propelled ground vehicles used to transport nuclear weapons in its basic structures. (Refer to paragraph 8.11 for design considerations of commercial vehicles).

a. Rated Load. The rated load is that load combination of forces (weapon and/or supporting equipment) that the basic equipment must support or resist in a static condition. This load is the nuclear certified load.

b. Design Loads. Design loads for each axis shall be based on the total static rated load multiplied by a factor of three or on the dynamic load multiplied by a factor or two, whichever is greater. To determine if the dynamic design load will exceed the static design load, the designer must consider lateral, horizontal, and vertical dynamic loads, acceleration encountered during ground transport environments, and shock loads associated with mate, demate, load, and unload operations. The design load shall be the minimum load for meeting the design stress.

c. Design Stress Levels. In determining allowable stresses for equipment, select the material and type stress specified in government publicatons and nationally recognized standards, such as MIL-HDBK-5, American Society of Testing Materials (ASTM), and Society of Automative Engineers (SAE). The combined stress level of the three axes shall be below the yield point.

8.2 CARGO AIRCRAFT SYSTEMS. Applicable noncombat delivery vehicles and support equipment as well as cargo aircraft carrying nuclear weapons must be designed to meet the general specifications for air transportability in MIL-A-8421, and the internal/external cargo handling requirements of the specified aircraft detail specification.

8.2.1 Cargo Aircraft Tiedowns. Nuclear cargo tiedown configurations shall:

a. Not impose reaction loads that exceed the strength of aircraft tiedown rings identified in the appropriate cargo aircraft.

b. Provide for only one tiedown device to one aircraft tiedown ring.

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c. Consist of symmetrical tiedown patterns parallel to the longitudinal axis of the aircraft.

d. Provide a minimum of four restraint devices on each item. Ten thousand-pound capacity chains, straps, and energy absorbing devices will be used as specified in the Cargo Loading and Airborne Weapons/Stores Loading Manuals for each aircraft, or as approved by the Naval Air Systems Command (NAVAIRSYSCOM).

e. Require each end of each tiedown device to terminate at a separate attachment point without going through any other attachment point.

8.2.2 Cargo Aircraft Tiedown, Palletized. Nuclear cargo tiedown configurations on pallets shall:

a. Not impose reaction loads that exceed the rated strength of the pallet tiedown rings.

b. Require one tiedown device to one pallet ring.

c. Consist of symmetrical tiedown patterns parallel to the longitudinal centerline of the aircraft, and within the restrictions defined for the pallet.

d. Require a minimum of four restraint devices on each item. Tiedown straps may be used for light items (250 pounds or less, total weight). However, for items over 250 pounds, 10,000-pound tiedown chains and assemblies which meet MIL-T-25959 or other restraints approved by the NAVAIRSYSCOM, will be used.

8.2.3 Cargo Restraint Criteria*

a. Determine the static design loads for the tiedown patterns by applying the minimum G load factors as shown in Table 8-1. These patterns will be structurally analyzed and/or

*NOTE: The listed criteria are minimum criteria based on nuclear weapons system safety considerations and may be made more restrictive by specific aircraft crash criteria or other factors.

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TABLE 8-1. Nuclear Weapon Tiedown G Load Factors for Cargo Aircraft.

LOAD DIRECTION	G LOAD	G LOAD
	COD Aircraft	All Other Aircraft
Forward	20.0	10.0
Aft	7.5	7.5
Lateral	3.0	3.0
Vertical (Down)	4.5	4.5
Vertical (Up)	10.0	3.7

proof tested, to ensure that they meet applicable G load factors before these patterns are published in the appropriate aircraft cargo loading manuals. The tiedown patterns may be verified in the form of calculated data. Off-pallet tiedowns which meet the criteria in paragraph 8.2.1, may be used to secure pallet loads in cargo aircraft.

b. In addition to the tiedown requirements of paragraphs 8.2.1 and 8.2.2, techniques or equipment should be developed that will help keep the nuclear weapons involved in aircraft accidents as close to the accident site as possible. (Examples are energy absorbing nets for aircraft rail and pallet systems and specialized energy absorbing containers for floor tiedown in cargo aircraft.)

c. Nuclear cargo will be placed forward of general cargo in the aircraft. The general cargo will be restrained to achieve a total minimim 10G forward restraint by the use of a restraining device (for example, a barrier net).

d. Handling and tiedown of the weapon shall comply with shipping restrictions for the weapon.

8.2.4 Aircraft Loading and Offloading Procedures. Loading or offloading procedures must meet the following criteria:

a. Steps must be in proper sequence.

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b. Positive measures must be taken to minimize weapon and equipment damage, such as:

(1) Placement of chocks to restrain ground handling equipment.

(2) Use of parking brakes.

(3) Use of manually operated safety devices.

(4) Use of observers to check clearances.

(5) Proper attachment of weapons or containers to handling or loading equipment or to the aircraft.

(6) Control of weapons or containers at all times.

8.3 CONTAINERS. Containers used for storage and transportation of nuclear weapons will incorporate design criteria specified in MIL-STD-209 and MIL-STD-648.

8.4 PALLETS.

a. Pallet design shall conform to MIL-STD-1366, and the cargo loading criteria of Table 8-1.

b. Pallets must have sufficient tiedown rings for the loads to be carried (paragraph 5.3.b.).

c. Pallet tiedown rings will be designed to meet test loads imposed by cargo load requirements (paragraph 8.2.2(1)).

8.5 TRAILERS. All free wheeling support equipment shall be designed to restrain the handling equipment and its rated load. All trailers used to transport nuclear weapons must meet the specific criteria in this paragraph and paragraph 8.1.

a. In a fully loaded configuration, trailer design will impede:

(1) Yaw, sway, or skid under operating and braking conditions and minimum stopping distances.

(2) Any tendency to tip, tilt, or jackknife.

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b. Parking brakes shall be designed to hold a fully loaded trailer on an 11.5-degree incline with the trailer headed up or down.

c. Trailers shall be equipped with an emergency breakaway feature to bring the trailer to a controlled stop in the event of disengagement from the tow vehicle. Semitrailer brakes shall be automatically applied in the event of breakaway from the tow vehicle. The design must allow for a manual override of the breakaway device when the trailer is being maneuvered by hand.

8.6 HOISTS, CRANES, AND SIMILAR DEVICES. The criteria outlined here apply to hoists, cranes, winches, and similar equipment. These must be designed, as a minimum to have:

a. Controls to ensure the load is under positive operator control.

b. Features to ensure prompt and automatic stop in case of operating mechanism failure.

c. Features to permit completely synchronized operations.

d. Features that will automatically neutralize the device in the absence of a manual force.

e. Maximum lift capacity and positioning limits and rates identified.

f. A minimum safety factor of three for all load sustaining (basic structure) parts, based on the yield point used for rated hoist capacity.

g. A minimum safety factor of five based on ultimate strength for load chains.

h. A minimum safety factor of ten based on ultimate strength for blocks and rope falls, fiber rope, and webbing.

i. A minimum safety factor of five based on ultimate strength for all accessory parts (such as hooks, rings, shackles, slings, and wire rope.

j. Hooks fitted with throat opening safety devices.

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k. Stops or limit switches to prevent overtravel of the hoist on rails and stop the chain or wire rope when the hook reaches its upper and lower limits.

8.7 FORKLIFTS. These criteria apply to the design of equipment used to lift nuclear weapons.

a. If more than one power actuating component in a mechanically parallel system is used in liflting the weapon, the components may be individually controlled to provide weapon attitude adjustments but must be capable of synchronization to provide a uniformly controlled lifting attitude.

b. The equipment shall be designed to prevent sudden dropping of the rated load in the event of hydraulic or electrical system failure.

c. Equipment shall be designed with pressure relief valves or regulators at appropriate locations in hydraulic and pneumatic systems to prevent overpressures. Internal leakage in hydraulic components of lift systems shall be limited so that the maximum drift rate is consistent at an operating rate that does not cause damage to the weapon.

d. Design of forklift and tines shall be adequate to safely meet all nuclear weapon operational requirements for the forklift. The forklift and rated load center of gravity must be compatible.

e. Forklifts shall include load attachment points to ensure positive control of the nuclear weapon at all times in the lifting and handling modes.

f. Mechanical stops or electrical switches shall be incorporated to prevent overtravel of forks in all degrees of control.

g. All forklift motion controls (except the parking brake, steering control, transmission, selectors, or similar devices) shall be self-centering. The engine start switch shall operate only when the automatic transmission is in the NEUTRAL or PARK position.

h. Parking brakes must be capable of holding a forklift with rated load on a 15-degree slope with the front of the truck pointed up.

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8.8 SELF-PROPELLED GROUND VEHICLES. This section covers self-propelled ground vehicles (trucks, vans, tugs, tractors, high-lift trucks, and other similar devices) used to transport nuclear weapons on its basic structure or tow (as a prime mover) other vehicles carrying the nuclear weapons. This section does not include loading/unloading trucks. Paragraph 8.11 applies for commercial equipment.

8.8.1 Tow Vehicles.

a. Controls must be designed to ensure towed vehicles are under positive control.

b. The brake system shall be functionally compatible with the towed vehicle brake system. The towing vehicle should not contribute to yaw, sway, skid, tilt, or jackknife under maximum braking conditions and minimum stopping distances.

c. The parking brakes, in combination with the towed vehicle parking brakes, must hold a fully loaded towing/towed vehicle combination, whether headed up or down on a 11.5-degree slope.

d. Vehicle connecting devices shall be compatible with that of a towed vehicle.

e. Fifth wheel tow vehicle configurations shall meet the safety latch requirement of MIL-M-8090.

f. Vehicle connecting devices shall have a minimum safety factor as stated in paragraph 8.1.2.c.

8.8.2 Self-Transport Vehicles. A vehicle used to transport nuclear weapons on its basic structure will:

a. Satisfy the criteria of paragraphs 8.1.1 and 8.1.2.

b. Have provisions for adequate load restraint.

c. Have parking brakes capable of holding the vehicle with rated load, while headed up or down, on a 11.5-degree slope.

8.9 WEAPON LOADERS. Weapon loaders include loading/lifting equipment, such as bomb lift trucks, high lift trucks, munition handling trailers with lifting devices, and loading/unloading trucks.

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a. The equipment shall be designed to allow safe control of the weapon in the event of electrical, hydraulic or pneumatic system failure.

b. The hydraulic and pneumatic systems must have pressure relief valves or regulators at appropriate locations to prevent overpressures. The maximum drift rate for weapon loaders used in weapon assembly and loading/unloading on combat aircraft shall be based on the required use of the device, but must not be enough to cause damage to nuclear weapon resources. The design goal is a maximum 0.5 inch per hour.

c. If more than one power actuating component in a mechanically parallel system is used to lift the weapon, the components may be individually controlled to provide weapon attitude adjustments, but must be capable of synchronization to provide a uniformly controlled lift attitude.

d. Design shall provide for positive control of the nuclear weapon at all times in the lifting and handling modes. One method is the use of load attachments and straps.

e. Mechanical stops or electrical switches shall be incorporated to prevent overtravel in all directions of the lift control.

f. Design shall include an "inching" capability in both reverse and forward directions.

g. Service and parking brakes must hold a fully loaded weapon loader on a 11.5-degree slope with the loader headed up or down.

h. All weapon loader movement controls (except the parking brake, steering control, transmission selectors, power takeoff, hydraulic pump, or similar devices) shall be self-centering. The engine start switch will operate only in the automatic transmission NEUTRAL or PARK positions.

8.10 GROUND TRANSPORT TIEDOWNS. The tiedown configuration for ground transport of all nuclear weapons must:

a. Provide the number of cargo tiedown points sufficient to restrain rated loads.

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b. Ensure the end of each tiedown device terminates at a separate attachment point without passing through any other attachment point.

c. Consider road conditions as identified in planned use documents.

8.11 COMMERCIAL VEHICLES. Structural design of commercial vehicles such as tow vehicles, fifth wheel, and forklifts, shall be based on appropriate standards, specifications, etc., to which the vehicle is manufactured. Commercial equipment shall meet applicable requirements of this document.

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SECTION 9

ELECTRICAL SUBSYSTEMS AND EMR HAZARDS

9.1 GENERAL. A major portion of any nuclear weapons system is composed of electrical subsystems designed to monitor, prearm, arm, launch, fire, and release nuclear weapons. These subsystems must be designed to preclude accidental operation, single circuit failure, or electrical disturbance from performing or degrading critical functions.

9.1.1 Electromagnetic Interference (EMI). All electronic and electrical subsystems/equipment in or associated with nuclear weapon systems shall be designed so that undesired responses and emissions are minimized. MIL-E-6051, MIL-STD-461, and MIL-STD-462 apply.

a. Wires, switches, cable connectors, junction points, and other system elements shall be designed to prevent undesired radiated and conducted interference or transients, when such EMI could result in a nuclear hazard or rocket motor ignition.

b. The nuclear weapon monitor and weapon control circuits, as well as release circuits, shall be protected from EMI. EMI protection must be carried through all applicable connectors and junction boxes.

9.1.2 Isolation. Isolation must be provided to prevent prearm, release, and consent functions from occurring as a result of a common malfunction. In addition, isolation is required to decrease the probability of activating critical functions during catastrophic events/abnormal environments. The following specific requirements apply:

a. Wire shields shall not be used as current carrying conductors. Shields will be covered by an insulation layer.

b. MIL-STD-1512 applies to electroexplosive circuitry affecting critical functions.

c. Circuits for nuclear consent functions shall be physically and electrically isolated from all other circuits and from each other.

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d. Prearming control circuits and safing circuits must be electrically isolated form all other circuits.

e. Prearming power circuits must be physically and electrically isolated from all other circuits; however, the power and control circuits may coincide at the final switching assembly prior to the nuclear weapon interface. Physical isolation within the final switching assembly shall be maintained to the maximum extent possible.

f. Monitor circuits must be electrically isolated from power and control circuits. Monitor functions shall be capable of being performed independent of weapon control functions.

g. All logic circuit signal leads must be located away from power leads and must be routed to prevent short circuits and minimize inductive and capacitive coupling.

h. For hard-wire systems, electrical functions which are peculiar to the aircraft nuclear weapon monitor, control, and release systems must not share an electrical connector with nonnuclear functions.

i. Electrical wiring, circuits, components, and so forth, employed, for functional operation of aircraft release or missile guidance must be electrically and physically iso- lated from the critical warhead circuitry.

j. Critical circuits must be isolated from potential sources of stray electrical power.

k. Hard-wire monitoring circuits shall be separated from functional circuits. Monitoring may be accomplished through separate circuits leading to back contacts or separate contacts of explosive switches, safing and arming devices, barometric switches, relays, and so forth.

9.1.3 AC Power

a. AC power circuits must have twisted leads with single point grounding of the return.

b. Structure return shall be avoided to eliminate common mode problems.

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9.1.4 Switching. Normally, the ground side of power circuits will not be switched. To meet two-fault safety criteria, ground switching may be used in addition to hot-side switching.

9.2 DESIGN CRITERIA FOR WIRING AND CABLING.

a. Wiring and cabling design must minimize coupling and provide for optimum separation within the available wiring space. Cable design must provide for adequate termination of shielded wires.

b. Cable construction must provide for twisting of each signal wire and the individual return wire. Equipment design and cable routing shall minimize circulation currents. Where employed, signal returns common to two or more circuits shall use a common ground reference connection.

c. Electrical wiring shall be installed and secured to minimize vibration and prevent chafing.

d. Except for weapon and warhead interface connectors, electrical power wiring must terminate in female connectors at the source outlet.

e. Wiring for critical circuits must be provided with mechanical support which is an integral part of the connector at the entry point into the electrical connector.

9.3 DESIGN CRITERIA FOR ELECTRICAL CONNECTORS.

a. Electrical connectors shall be designed to prevent misalignment of connectors, minimize bent pins during mating, and prevent mating of adjacent connectors. Only one wire per pin will be used. Connector pins shall not be used as a terminal or tie-point for multiple connections. Spare pins shall not be used for mechanical support of in-line components. (Such practices result in an inherent multi-fault condition.)

b. All hard-wire electrical connectors associated with aircraft nuclear weapon circuits shall conform to MIL-C-38999 (for critical circuits). (Electrical connectors used within armament system black boxes are exempt from this requirement.) Potting compounds, if used in electrical connectors, must positively preclude reversion.

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c. The assignment of functions to pins within a connector shall ensure that a short circuit occurring as a result of a bent or misaligned pin shall not result in delivery of more than 100 milliamperes current to the nuclear weapon or to critical function circuits.

d. Connectors used to carry shielded wire shall use a back sheel that provides for peripheral bonding of the shield. Nonconductive finishes which would limit the bonding of the shields shall not be used on these connectors.

e. Connectors shall not contain both critical function circuits and liquid lines.

9.4 ELECTRICAL CURRENT CONSIDERATIONS.

a. Monitoring and testing current must be limited to a functional value below the level required to actuate the most sensitive component in the nuclear weapon system. Currents in the nuclear weapon/warhead monitoring and continuity, EEDs, and pressure testing circuits must not exceed 100 milliamperes.

9.5 DESIGN CRITERIA FOR PANEL CONSTRUCTION AND PACKAGING.

a. Panel construction for critical electrical functions, if not hermetically sealed, must have all terminals fully insulated (or the panel fully enclosed) so that foreign objects cannot enter to cause terminal-to-terminal or terminal-to-ground short circuits. If a control panel applies power to the nuclear weapon, and is not hermetically sealed, it must be fully enclosed and all electrical contacts insulated. (Drain holes required by applicable military specifications are permissible.)

b. Logic elements used for the control, monitor, and release of nuclear weapons must be fully insulated if not hermetically sealed to prevent terminal-to-terminal or terminal-to-ground short circuits caused by moisture or foreign objects. The element may be a plug-in unit which is a part of a larger assembly. In such cases, the design criteria in paragragh a. above applies.

9.6 EMR HAZARDS TO NUCLEAR WEAPON SYSTEMS. Design of nuclear weapon systems shall consider the inherent susceptibility of EEDs and semiconductor devices to EMR.

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9.7 DESIGN TO ELIMINATE OR MINIMIZE EMR EFFECTS.

9.7.1 General EMR Hazard Protection. Maximum practicable protection is required against the hazards of EMR, including lightning and EMP. The shielding design philosophy employed must recognize that the entry of electromagnetic energy into a system can occur through cable and wire coupling, windows and deliberate antennas, and the weapon system (acting as an antenna).

9.7.2 Shield Design. The following rules for achieving satisfactory shielding design apply:

a. Emplace all susceptible components, subsystems, and systems in metallic enclosures (for example, storage and transport containers for missiles, shells, and reentry vehicles provide considerable protection against electromagnetic energy).

b. Provide radio frequency (RF) shielding for openings which provide access to cables, connectors, and other electrical equipment.

c. Provide RF shielding for cables attached to susceptible components and for all cables attached externally to the nuclear weapon.

d. Provide attenuators, filter networks, lightning arrestor connectors, or electrical surge arrestors where practicable.

e. Use circuit-isolation relays or switches which are insensitive to RF energy and position them as close as possible to the elements to be protected to prevent energy pickup by long leads.

f. Ensure all section joints of the weapon are low-resistance contacts so that the weapon skin serves as an RF shield.

g. EED firing circuits shall consist of two wires, twisted together and shielded. A single ground point shall be common to all firing circuits.

h. Design the system such that, regardless of the no-fire threshold of the EED, the maximum Root Mean Square (RMS) current experienced in the bridgewire shall be 20 decibels below the

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maximum no-fire current when exposed to the stray electrical energy environment specified in the STS or 100 watts/meter^2 average whichever is greater.

i. Monitor circuits shall be designed so that they will not conduct or couple EMR energy into functional or control circuits.

j. The case of an initiator shall be electrically bonded to the structure. All removable parts of the case, which are removed and replaced during operational and maintenance procedures, must not degrade the shielding after replacement or reassembly.

k. Maintain electrical continuity and isolation of circuitry shielding. Shields shall not be used as intentional current carrying conductors. Metallic (water-proofed) shielding caps shall be provided and shall ensure electrical continuity from shield to case with no gaps or discontinuities in the shielding configuration. When EED male connectors are in an open circuited configuration, shielding caps shall be installed.

9.8 EMR ENVIRONMENT THROUGHOUT THE STOCKPILE-TO-TARGET SEQUENCE

a. A comprehensive survey of the EMR environment which weapons will encounter is not available to date. However, the Naval Surface Weapons Center (NSWC), Dahlgren Laboratory has measured the EMR levels existing at selected locations at various Navy bases in the CONUS and on board various classes of Navy ships.

b. A general conclusion drawn from the NSWC measurements is that the ground-based, shipboard, and airborne radio transmitters now in the Navy inventory generate average field intensities ranging from a few microvolts per meter to about 300 volts per meter.

c. Radar transmissions differ from continuous wave communications transmissions in that RF energy is emitted as a series of pulses (lasting but microseconds) interspersed with periods of radar silence, during which the return "echo" is awaited. The power transmitted in one of these short pulses is called the peak power. The NSWC surveys indicate that radars can generate peak power densities of about 75 dBm/m^2 in the near vicinity of the antenna.

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d. The above mentioned power densities are presented here as an indication of the order of magnitude of the EMR environment anticipated. The environment actually experienced by a particular weapon throughout the STS can be determined by the application of well-known antenna theory, once the location and characteristics of all RF emitters in the vicinity of the weapon have been determined.

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SECTION 10

TEST AND TRAINING EQUIPMENT

10.1 GENERAL. The design criteria outlined here apply to test equipment and training devices for combat aircraft and missile/rocket systems. A nuclear bomb or warhead shall not be used as a test or training device.

10.2 TEST EQUIPMENT.

a. Test equipment shall be designed to:

(1) Preclude malfunctions in the test equipment or the test circuits which could operate critical nuclear weapon functions.

(2) Preclude faults within a tester that could degrade the nuclear safety of the equipment to be tested.

(3) Minimize the introduction of stray voltage, non-limited current, or EMR energy into the weapons.

b. Test equipment must ensure that any weapon system component which has been operated during testing is in the safe or inactivated position at the completion of the test. The safe position of such components shall be verified by a positive indication.

c. Equipment used to test the control, launch, or release systems shall meet these design criteria:

(1) Test equipment shall be used only where necessary to establish and verify system operation, reliability, and safety. The majority of any required testing shall be done before the nuclear weapon is mated to the combat delivery vehicle.

(2) Intervals between required tests on nuclear weapons or weapon systems shall be the maximum interval needed to maintain a high confidence level in the system operation and safety.

d. The test equipment design agency shall provide periodic maintenance, inspection, and test requirements for the test equipment so that when applied, the test equipment will continue to meet the original specification.

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e. Based on system requirements, test equipment shall be designed to reveal:

(1) Improper wiring

(2) Line-to-line shorts

(3) Line-to-ground shorts

(4) Stray voltage

(5) Improper system operation

f. When nuclear weapons are present, electrical test equipment, including built-in test equipment (BITE) must be incapable of:

(1) Generating release or launch signals.

(2) Causing any critical nuclear function to occur.

(3) Negating two-man control for release or prearming.

(4) Unlocking the in-flight operable bomb rack lock.

(5) Activating an A/D or S/A device.

(6) A single component failure causing any of the actions addressed in paragraphs f.(1) and (5) above.

g. Electrical testing which uses separate ground support equipment (GSE) must accomplish the following:

(1) Preloading tests must reveal faults in any of the critical functions. Specifically, criteria of paragraph 5.1 must be met.

(2) Preflight testing, postilight testing, and periodic testing must be accomplished at the farthest downstream interface connectors of the nuclear functions to be tested. Control panel connectors, bulkhead connectors, and other upstream connectors need not be disconnected for routinely scheduled tests.

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(3) Isolation resistance tests shall be accomplished wherever possible during routine time phase testing mentioned in paragraph c.(2) above.

(4) Isolation resistance tests of all nuclear circuits shall be accomplished on a periodic basis.

(5) If applicable, testing shall be accomplished to interface requirements stated in the AMAC specification.

h. Testers shall not be capable of appyling sufficient power to cause the operation or firing of any item being tested unless specifically designed to do so.

i. Automata and software used for testing or checkout must meet the provisions of Section 6 of this Handbook.

j. Interference and susceptibility within test equipment shall be controlled by adequate provisions to eliminate undesired responses and emissions from all electronic and electrical equipment in, or associated with, nuclear weapon systems.

k. Wires, switches, cable connectors, junction points, and other electrical system elements, as appropriate, shall be designed to prevent undesired radiated and conducted interferences or transients.

10.3 TRAINING DEVICES. Training devices or nuclear weapon simulators shall not be used with nuclear weapons. These items do not require certification action; however, those items used to duplicate a weapon or a weapon system function must have an explicit means of identification as a training item.

a. Training devices incorporating explosive-actuated devices (for example, parachute deployment mechanisms) must be designed to ensure that a single component failure will not actuate the device.

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SECTION 11

ADDITIONAL DESIGN CRITERIA

11.1 GENERAL. The criteria outlined in this section are broad, which are important in the overall design process, and apply to all designs discussed in this Handbook. Evaluation of these criteria is implicit in the examination of the specific systems.

11.2 HUMAN ERRORS. Using accepted human engineering methods, weapon system designers shall conduct human error analyses and error reduction studies to identify system modes which can result in hazardous conditions. Based on the results of these analyses, features to either eliminate human error or limit the consequences of human error shall be incorporated.

a. At least two human errors, independent in nature, must be possible without initiating a sequence of events which will result in a significant nuclear yield. Features must be incorporated to minimize the number of points within the system where human acts could degrade nuclear safety.

b. Positive measures must be developed to prevent accidental actuation of critical software commands or hardware controls when it may degrade nuclear safety. Specifically, commands, controls, and switches directly affecting the safe conditions of nuclear weapons shall:

(1) Be a different design (shape and mode of operation) from any other immediately adjacent controls and switches, or in the case of operator initiated software commands, the commands must be distinctly different from any other commands used and must consist of several characters spaced widely apart on the operator keyboard so as to prevent accidental or inadvertent generation of the unique commands.

(2) Require two or more distinct movements before they are activated (such as inserting a key and then turning the key assembly, or raising a guard and then moving a toggle switch, or other similar methods).

(3) Incorporate a device to prevent the switch, which singularly controls or selects nuclear weapons and conventional weapons (such as a master armament selector switch), from being placed accidentally in the nuclear weapon position.

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11.3 PROTECTION OF FRIENDLY TERRITORY. Weapon systems shall be designed and controlled to prevent nuclear detonations except within specified selectable boundaries.

a. Guided missiles launched from aircraft must receive a good guidance signal from the guidance and control unit before final arming of the nuclear warhead can occur. The good guidance signal, designed into system control logic, shall be withheld if a final guidance accuracy check indicates that impact will occur outside specified target boundaries.

b. The boundaries for airborne release and delivery systems vary with the number of weapons, weapon yield, type, method of employment, geographical location, and operational requirements. The crew of the aircraft has direct control of nonguided weapons. Therefore, they must exercise positive control over the safety features of the weapon system in order to preclude nuclear detonations outside specified boundaries (for example, in the event of inflight accidents, intentional destruction actions, aborted missions, or weapon jettisoning).

11.4 EXPLOSIVE ORDNANCE DISPOSAL (EOD). Aircraft and missile systems must permit emergency access to those components and circuits required to carry out render-safe procedures.

11.5 SECURITY.

a. Consistent with operational requirements, system safety design must incorporate features and involve techniques that will contribute to adequate security. Weapon system design must provide for physical security control compatibility such as command disablement, dimensional sizing for secure structures, etc., and for access denial and security force action.

b. As a minimum, a security system must prevent undetected access to a nuclear weapon and provide protection of nuclear safety critical equipment or secure data.

c. The design must ensure security control and protection of nuclear safety critical cards, tapes, keyboards, input controls, components, and devices which are developed to control or command the coding, enabling, prearming, arming, firing, launching, or releasing of a nuclear weapon(s)/weapon system.

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11.6 ENVIRONMENTAL PARAMETERS. Nuclear safety design features must be considered over the full range of credible normal and abnormal environments which the system could experience. Specific normal and abnormal environmental parameters are system dependent and will be specified in the bomb/warhead stockpile-to-target sequence document and in carrier and/or missile environmental specifications.

11.7 MULTIPLEXED MONITOR, CONTROL, AND RELEASE SYSTEMS (MMCRS). The following criteria must be met for all aircraft monitor and control/release (AMAC/REL) functions in multiplexing systems:

a. Time division multiplexing (TDM) shall be required for nuclear AMAC and release systems.

b. MIL-STD-1553 will apply for standardization purposes.

c. When two-man controlled safing switches are incorporated, the activation of either switch results in the application of safing power to the weapon.

d. In any multiplex stores management system, hard-wire discrete consent functions are required for weapon prearming and unlocking the in-flight operable bomb rack lock to prevent these functions from occurring even though the command for such functions has been transmitted over the multiplex bus. Circuits for these consent functions shall be physically and electrically isolated from all other circuits and from each other (Section 5.1.(c)(5)). An exception to these hard-wire consent function requirements may be granted where adequate statistical proof is offered to assure that the likelihood of inadvertently prearming the weapon or unlocking the in-flight operable bomb rack lock is less than 1 in 10^{10} during the lifetime of the weapon system or is less than 1 in 10^{7} for any occurrence of a catastrophic event/abnormal environment.

e. No single component failure, logic failure, or system fault shall allow inadvertent activation of control signals which prearm or effect separation; for example, rack unlock, release, launch, eject, fire, of a nuclear weapon. This requirement is independent of hardwired consent functions (paragraph c. above).

f. No single component failure or system fault will prevent weapon safing capability.

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g. A single data transfer action which affects the state of a prearm or separation control signal must be discriminated from other multiplexed data, or the effects of a maximum duration noise burst, which could be interpreted as a data function commanding the generation of a prearm or separation control signal. This required discrimination may be accomplished by increasing the number of bits, having a check at the receiver for one of a limited number of acceptable signals (excluding PAL), or establishing a range of valid voltage amplitudes. This range would vary from a threshold voltage (below which is rejected as noise) and a maximum peak-to-peak voltage which is less than normal aircraft line voltages (for example, 28 volts).

h. A sufficient number of transmissions must be employed to ensure that a single noise burst will not result in the generation of uncommanded prearm or separation control signals.

11.7.1 MMCRS Power.

a. The MMCRS design shall permit the operator to initiate the application or removal of power from any MMCRS unit which interfaces directly with a nuclear weapon interface (NWI).

b. Weapon control power shall be supplied to the switching assemblies ONLY when the MMCRS operator has given consent to safe or prearm the weapon, unlock the releasing mechanism, or release the weapon.

c. Application, Removal, and Loss of Power. Applying power to, or removing power from, any portion of the MMCRS shall not result in the generation of false or uncommanded output signals.

(1) Start-up Routines. After application of power to a station, self tests must be performed before the station is capable of outputs to the weapon or release system.

(2) Shutdown. Each multiple station shall have the ability to shut down independently of all other stations and central control functions. The shutdown sequence shall be accomplished in a manner so that no switching of redundant circuitry and channels occurs.

(3) Power Dropout. Each multiplex station shall operate in a safe manner (proceed in the normal shutdown routine) during any dropout interval of the vehicle power system.

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11.7.2 Protection Against Abnormal Environments. The following design requirements apply to provide maximum nuclear safety in abnormal environments.

a. Maximum physical separation of parts, modules, assemblies, etc., shall be provided with MMCRS units to prevent such items from degrading nuclear safety by physical contact under catastrophic conditions.

b. Maximum efforts will be made to keep nonrelated and dissimilar MMCRS functions (for example, safe and prearm) physically separated and isolated within each MMCRS unit.

c. Voltage and current levels required for operation of the remote multiplexing terminal (RMT) will minimize the probability of affecting weapons or release mechanism operation if these voltages and currents were applied to the critical circuits at the nuclear weapon interface because of a catastrophic event.

11.7.3 Prearm and Safe Signal Functions. Prearm and safe signal functions (command, activate, control) will not be used as complementary functions. Example: Prearm signal ON state will not be used as safe signal OFF state or vice versa.

11.7.4 MMCRS Release Signal. The MMCRS shall inhibit the generation of a release signal until a rack unlock signal has been generated to the weapon rack and confirmation is obtained by monitoring the rack locking system.

11.7.5 MMCRS Control. The operation of the MMCRS shall be under control of the launch vehicle crew at all times. The system must provide the crew with both general and specific selection of critical functions, so that flexibility is optimized.

11.7.6 Fail Safe Hardware Design. All multiplexer elements shall assure the probability that all critical functions will fail to a safe state. The safe state is defined as the absence of all power at the nuclear weapon interface.

11.7.7 MMCRS Remote Terminal. An electrical fault to the input of a remote terminal (RMT) shall not impair other MMCRS functions.

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11.7.8 MMCRS Memory Elements. Memory elements used to store MMCRS operational programs and algorithms for MMCRS operational control and AMAC/REL logic processing shall not require electrical power for retention of data, shall be of the nondestructive readout type, and shall not be alterable except by deliberate, manually controlled action.

a. If the MMCRS operational program and AMAC/REL logic processing routines are stored in a common memory element that is shared with operations other than the AMAC/REL functions, special precautions shall be taken to isolate the AMAC/REL section and prevent inadvertent and unauthorized entry and modifications. Special hardware or software implementation shall be provided to ensure that this requirement is met. This same requirement shall be applied to separate dedicated memory elements which are used only for AMAC/REL functions.

11.7.9 MMCRS/RMT Switching. The MMCRS and/or RMT must provide break-before-make action between changes of state of all control signals presented to the nuclear weapon interface.

11.7.10 MMCRS System Test Requirements.

a. In-flight self testing, if provided, shall not interfere with normal multiplexer operation. In no instance shall the in-flight testing cause the generation of control signals at the nuclear weapon interface or degrade nuclear safety.

b. Provisions for ground testing shall be designed so as not to degrade nuclear safety.

11.7.11 MMCRS Additional Requirements. The requirements of Section 9 apply to MMCRS systems.

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SECTION 12
NUCLEAR SAFETY EVALUATION CRITERIA

12.1 GENERAL. This section outlines the general evaluation criteria which must be applied to combat aircraft, missiles, noncombat delivery vehicles, support equipment, and subsystems associated with nuclear weapons prior to their certification.

a. The Navy organization or contractor responsible for each specific system or subsystem must develop the specific plans, tests, and analyses necessary to demonstrate the operation of nuclear safety features. The baseline for applying evaluation criteria is drawn from user requirements. Formal documents (such as Operational Concepts, Stockpile-to-Target Sequence (STS), and the Military Characteristics (MCs) for nuclear weapon development) are the primary sources for these requirements.

b. Test, analysis, and evaluation require the exercise of sound judgment by personnel responsible for nuclear safety in all phases of system acquisition, management, modification, and operation.

(1) The data derived from tests and analyses are essential to accomplishing nuclear safety evaluations. Both qualitative and quantitative analyses provide a basis for the nuclear safety evaluation.

(2) Specific criteria for an item shall be reviewed for applicability and used for subsequent evaluations necessitated by retrofit changes. New uses or features for equipment shall necessitate review of specific criteria.

(3) A description and the results of tests and analyses performed on each item shall be provided according to MIL-STD-882 and OD 44942.

c. When Military Specifications or Standards which satisfy nuclear safety requirements exist, test and analysis requirements may be satisfied by demonstrating accomplishment of those specifications or standards. When operational requirements are found to deviate significantly from the test requirements indicated in the Military Specifications or Standards, the test requirements shall be supplemented accordingly.

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d. Those safety features, subsystems, or subassemblies of each item of equipment having an affect on nuclear safety shall be identified.

12.2 ENVIRONMENTAL CRITERIA. The environmental requirements for each nuclear weapon system and associated equipment must be derived from the sources below for the predicted operational environment.

12.2.1 Nuclear Weapon Systems. The basic document used to determine operational environments, both normal and abnormal, is the STS. Another prime source of environmental data is the ERDA/DOD Environmental Data Bank.

12.2.2 Equipment. The testing requirements for support equipment will generally be derived from the STS and other equipment Military Standards. Equipment testing shall be performed to actual environmental levels when these levels exceed applicable MIL-STD requirements.

12.3 SAFETY REQUIREMENTS. Safety requirements must be met in the following order:

a. Safety shall be assured through appropriate design features.

b. When safety requirements cannot be met by basic design, appropriate additional safety devices shall be used.

c. When the required degree of safety cannot be ensured through design or additional safety devices, special operational or administrative restrictions shall be developed for the system.

12.4 COMBAT AIRCRAFT SYSTEMS. This section deals with nuclear safety evaluation of aircraft delivery and release systems.

a. Evaluation requirements and procedures for obtaining nuclear safety approval of a manned nuclear weapon system consist of an evaluation of the design by analysis and test (where applicable) and a study and physical examination of the weapon system in accordance with OPNAVINST 8110.20. The primary requirements for nuclear safety are outlined in the applicable design criteria sections (see paragraphs 5.1, 5.4, 5.5, 11.1, and 11.2).

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b. Test, analysis, and physical examination will provide assurance that nuclear safety criteria have been met.

(1) Analysis consists of a nuclear system safety analysis based on the STS, supported as necessary by other specialized analyses.

(2) Fault tree type logic, when used as an analysis technique, need not be formalized or quantified unless directed by the responsible Navy agency. Informal examination and analysis shall be performed continuously by the responsible Navy agency as technical interchanges begin.

(3) Participation in acquisition phase milestones (for example, Preliminary Design Reviews (PDRs), Critical Design Reviews (CDRs), Final Design Reviews (FDRs), and First Production Units (FPUs)) are part of the informal analysis procedures.

(4) A study of the AMAC Interface is to be considered a part of the nuclear safety analysis.

12.4.1 Additional Requirements (When Specified).

a. A review and listing of all assemblies, wires, connectors,
and other hardware which, when considered totally, define the AMAC and nuclear suspension and release systems. This analysis defines the nuclear configuration.

b. An analysis of interrelationships between the nuclear weapon system and other systems of the aircraft.

c. Bent pin analyses performed on connectors which transmit critical nuclear functions.

d. EMR compatibility analysis.

e. Ground equipment (paragraphs 12.8, 12.11, 12.12, 12.13, and 12.16) and built-in test equipment analysis.

f. Failure mode and hazardous effects analysis of nuclear configuration components and subsystems whose operation or failure could cause critical nuclear functions to occur. These functions are defined as:

(1) Power to the aircraft nuclear weapon interface.

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(2) Circumvention (deliberate or accidental) of two-man control for prearming and release or launch.

(3) Operation of the in-flight operable bomb rack lock (IFOBRL).

(4) Accidental separation of a nuclear weapon from the aircraft.

g. Preliminary hazard analysis to identify potential or real hazards to the nuclear configuration or nuclear weapon as a result of a nonnuclear system operation or failure.

h. Logistic support analysis to assure that maintenance philosophy and procedures are safe.

i. Analysis to ensure that automata requirements have been met (paragraph 12.5).

12.4.2 Tests

a. Using production fleet-configured aircraft and test load devices, demonstrate the capability of the AMAC system to meet design requirements: voltage, current, switching characteristics, and worst load conditions defined in the AMAC specifications.

b. Conduct circuit isolation tests to ensure that the nuclear configuration meets design criteria for electrical isolation between unlock, release, launch, AMAC circuits, and all other electrical circuits.

c. Conduct EMR and EMI tests to ensure that onboard emitters and switching functions cannot cause critical nuclear safety functions to occur.

d. Conduct functional testing of the nuclear system on production aircraft.

12.5 NUCLEAR WEAPON SYSTEM AUTOMATA. Automata shall be nuclear safety certified. Automata software is designated for a Software Nuclear Safety Analysis (SNSA) by the NWSSG, when convened, or AIR-54133.

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12.5.1 Hardware Evaluation Requirements. To ensure automata nuclear safety standards are met, the following nuclear safety design areas shall be checked for proper operation by the NWSSG.

a. Program loading and memory data change.

(1) Detection of errors and operator notification.

(2) Reset or synchronization functions can be engaged.

(3) Automata operation is precluded until termination of proper sequence (this may be software controlled).

(4) Reloading or changing of critical memory contents is precluded unless the designated means for entry is properly exercised (this may be software controlled).

b. Control.

(1) Self-test routines must be demonstrable. In lieu of such self-test capability, the automata must demonstrate automatic shutdown upon erroneous or incomplete functioning (this may be software controlled).

(2) If the equipment interfaces with a human operator, such ancillary equipment as necessary for control shall be checked for the extent of control it exerts on the weapon system and for its susceptibility to control by unauthorized persons.

(3) Failure modes must be analyzed to assure that they cannot contribute to unauthorized operation of weapon system critical functions.

c. Security.

(1) Security of data contents must be assured through software controls. The memory must accommodate erasure, overwrite, or simple destruction in order to preclude divulgence of classified contents. Such a characteristic must be clearly demonstrated.

(2) An evaluation must provide assurance that the hardware meets current National Security Agency TEMPEST requirements. Functional areas which do not process classified information are exempt from this criterion.

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12.5.2 Hardware Data Requirements. The engineering group responsible for system design or modification shall review all data pertinent to nuclear safety and provide the applicable data to the agency responsible for preparing the Technical Nuclear Safety Analysis for review by the Nuclear Weapons Safety Study Group (NWSSG). It must be determined to what extent an automatic or semiautomatic nuclear weapon system is susceptible to unauthorized or accidental actuation. The susceptibility analysis shall be summarized in the draft Nuclear Weapons Safety Study Report (NWSSR) prepared by the Naval Weapons Evaluation Facility (NWEF).

12.5.3 Analysis Data Requirements. Analysis data shall be provided as defined in paragraph 12.16.

12.5.4 Software Data Requirements. At the earliest possible date, the Program Manager will make preparations to analyze new or updated software for existing nuclear weapon system requirements. Questions pertaining to the requirement to perform SNSA on a particular piece of software should be addressed to AIR-54133. However, a SNSA shall be required for those software items which are critical to the release/launching and/or operation of the nuclear weapon.

12.6 REQUIREMENTS FOR ARMING AND FUZING (A&F) SYSTEMS. All arming and fuzing systems shall be evaluated at the system and component levels through analysis and tests to assure nuclear safety during logistic movement, storage, and maintenance within the STS. Specific requirements are:

a. Identify each critical component whose malfunction or failure could contribute to premature prearming or arming.

b. Identify the failure mode of each critical component that contributes to premature prearming or arming.

c. Assign a likelihood of failure to each critical component failure mode.

d. Include a fault tree analysis of all critical component failure modes that could contribute to premature prearming or arming.

e. Include a nuclear safety analysis of the system for both normal and abnormal environments which, as a minimum determines the likelihood of:

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(1) Premature operation during storage and logistic operations (system not prearmed).

(2) Premature operation for each stage of prearming.

(3) Premature firing after the system is fully armed.

(4) Premature operation, as the result of abnormal environment exposure as defined in the applicable STS.

f. Thermal analyses with a graphical "temperature-time" presentation showing relative component temperature as a function of exposure time. Thermal analyses must cover every component designed to provide thermal protection that could affect critical functions, and all explosive components.

g. Graphic presentations of all-operate and no-operate response characteristics for each normal and abnormal environmental sensing device.

h. Bent pin and connector mismating and misalignment analyses for each critical component connector to ensure against accidental prearming.

i Cable routing analysis to ensure against cable damage during maintenance operations which could contribute to premature prearming or arming.

12.7 GENERAL EVALUATION OF NONCOMBAT DELIVERY VEHICLES AND SUPPORT EQUIPMENT. Evaluation requirements for certification consist of analysis, examination, or testing by the responsible Navy agency.

a. If fail-safe features are incorporated, they must be evaluated or tested to determine that they provide safe control of the weapon in any system failure. The weapon may not be supported by lift arms, cables, or hydraulic system, but must be supported by the basic frame of the equipment during both air and ground transport. (This requirement does not apply to equipment used solely to position or transfer weapons.)

b. The equipment must prevent or reduce the transmission of fire to nuclear weapons. This requirement can be demonstrated by analysis, testing, or a combination of both.

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c. The equipment must meet minimum roadability requirements as specified in the STS concept of operations, and logistic handling.

d. First Article Analyses and Structural Tests. These analyses or tests determine that the design factors of paragraph 8.1.2 have been met. (For wheeled vehicle testing, the article is to be supported only by the structural members since it is not intended that tires withstand required test loads.) Adequacy of the designated structural safety design factor (a minimum of three times the static rated load or two times the dynamic load, whichever is larger in each axis), shall be proven by one or more of the following ways:

(1) Analysis. Performance of a detailed stress analysis. This analysis may be supplemented by selective structural tests.

(2) Nondestructive Tests. Testing to conditions which approach the yield strength of the structure with suitable instrumentation at all critical stress points as determined by stress analysis. These tests should not cause yielding of any component. (Yield has occurred when strain instrumentation reveals a permanent set, as defined by material properties.) Results of these tests shall be combined with an abbreviated stress analysis to determine whether the yield meets the required design factors. The test design load must be determined by analysis, and must include the static design load or appropriate dynamic design load (whichever is greater in each axis). Mobility test results will be used to determine the total dynamic load. These test loads must be applied simultaneously to the test vehicle along the appropriate axes.

(3) Destructive Tests. Test loads shall be applied simultaneously to the test vehicle along the appropriate axes until the item fails catastrophically. The test loads at this point must be related to a load at the material yield point greater than the static design load or the dynamic design load.

e. Operational Vehicle Proof Tests. Tests shall be performed on at least one fully configured article and other designated samples as necessary to determine that the item will withstand specified rated loads. Example methods are:

(1) Testing the first fully configured article to 110 percent of rated load and all other articles to 100 percent of rated load.

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(2) Testing the first fully configured article and selected samples to 110 percent of rated load.

f. Rated load and gross vehicle weight shall be clearly identified on all vehicles.

12.8 CARGO AIRCRAFT SYSTEMS.

12.8.1 Cargo Aircraft Tiedowns. All equipment used for the transport of nuclear weapons in cargo aircraft shall have developed tiedown configurations.

a. Tiedowns must satisfy design criteria in paragraphs 8.2.1 or 8.2.2.

b. Tiedown patterns shall be structurally tested or verified by analysis according to G load factors of Table 8-1. Pallet loads, when tested, will be secured to a simulated aircraft system.

12.8.2 Tiedown Tests or Analyses (Paragraph 8.2.3). Simulated loads shall be applied according to Table 8-1 in the development of tiedown patterns. A tiedown pattern is considered acceptable if the rated strengths of the aircraft floor and pallet fittings are not exceeded; the rated strengths of the restraint devices are not exceeded; and the total vector load on any tiedown device attached to the test item does not exceed its rated strength. During application of simulated loads, a determination shall be made on required use of additional aids (such as under the axle or under the frame shoring) for equipment with flexible suspension systems and pneumatic tires.

12.9 REUSABLE CONTAINERS. The acceptance criteria contained in MIL-STD-648 and MIL-STD-209 apply.

12.10 HOISTS, CRANES, AND SIMILAR DEVICES.

a. The procurement/development agency and the nuclear safety evaluation agency shall determine if the design fsctors have been met and will identify additional tests if necessary.

b. All hoists shall be tested to 125 percent of the rated capacity.

c. Test results at rated load shall be used to determine effectiveness of the fail-safe and synchronization features.

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d. Electrical, hydraulic, and pneumatic systems shall be analyzed for safe operation at rated capacity.

e. Safety devices (for example; braking devices, locking pawls, safety catches, rail stops, limit switches, and valves) shall be tested at rated load.

f. The procurement/development agency and nuclear safety evaluation agency shall determine the required environmental tests.

12.11 FORKLIFTS AND WEAPON LOADERS.

a. The first article and proof load test and system failure test requirements of paragraph 12.7 shall be met.

b. Test all lifting devices at test site ambient temperatures and then repeat at the extreme operating temperatures using rated load capacity.

c. Conduct drift rate tests with the lifting device positioned at extreme and midheights. Internal leakage in hydraulic components of lift systems must be limited so that the maximum drift rate is consistent with the precision required to prevent damage to a weapon. The design goal for weapon loaders used in assembly and loading and unloading combat aircraft is a maximum drift of 0.5 inch per hour.

d. If more than one power actuating component in a mechanically parallel system is used in lifting the weapon, the component may be individually controlled to provide for weapon attitude adjustments and must be capable of synchronization for a uniformly controlled lifting attitude.

e. Determine that hydraulic and pneumatic systems are designed to prevent overpressure.

f. Analyze or test any attachment or lifting device fitted to a vehicle for compatibility in the worst case environment to which the vehicle will be subjected.

g. Test the parking brakes to ensure that a fully loaded vehicle does not roll on slopes as specified in paragraphs 8.7.h. and 8.9.g.

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12.12 COMMERCIAL VEHICLES. Equipment shall be evaluated for nuclear safety adequacy in accordance with appropriate standards, specifications, designated tests, etc. Commercial equipment must meet the requirements of this document to the extent of its use with nuclear weapons. The responsible evaluation agency must assure that nuclear safety is not compromised for the intended use.

12.13 TRAILERS, TOW VEHICLES, AND SELF-PROPELLED VEHICLES. Test the service brakes of all vehicles including tow vehicles and towed vehicle combination brake systems, as applicable.

12.13.1 Service Brakes. Test the service brakes of all vehicles by progressively increasing the speed from which stops are made, in increments of approximately 5 mph, up to the maximum rated speed of the vehicle, until failure occurs (or until the maximum safe speed has been attained, whichever occurs first). Make the initial tests on a dry, brushed, level, concrete surface. Stop the vehicles by operating the brake system to produce maximum braking force (that is, "panic stop" conditions). Repeat the procedure on surfaces similar to those expected during the operational life of the vehicle. In each test, determine the maximum safe towing speed and record the following data on the brake performance as a minimum:

a. Tendency to yaw, sway, or skid.

b. Tendency to tilt, turn over, or jackknife.

c. Damage or excessive wear.

d. Stopping distances.

e. Towing speed.

f. Contact of wheels with ground.

12.13.2 Accidental Trailer Towbar Disengagement. Test the emergency braking system of trailers using towbars. Accidental trailer towbar disengagement may be verified by full scale testing or by analysis and limited testing. Limited testing may be specified by the procurement/development agency and nuclear safety evaluation agency. For full scale testing, the following criteria apply: While towing the fully loaded trailer at maximum expected operating speeds over a straight, smoothly

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paved road, disengage the towbar from the tow vehicle, and watch the emergency braking action of the trailer. The following items must be considered to determine whether the emergency brake system is adequate.

a. Distance from point of towbar disengagement to final stop.

b. Lateral distance of travel from point of towbar disengagement to final stop.

c. Attitude of trailer at time of stop.

d. Damage incurred by trailer and/or load as a result of disengagement.

12.13.3 Parking Brakes. Test or analyze the parking brakes to ensure that the fully loaded vehicle does not roll when parked on an 11.5-degree slope. Conduct the test with both the forward and aft end of the trailer up the slope. The procurement/development agency and nuclear safety evaluation agency will specify if analysis is to be used in lieu of testing.

12.13.4 Associated Equipment. Test all equipment used to carry nuclear weapons on vehicle basic structure against the criteria of paragraphs 12.7.e. and 12.7.f.

12.13.5 Mobility Requirements. The procurement/development agency and the nuclear safety evaluation agency shall determine the applicable mobility requirements of MIL-M-8090.

12.14 GROUND TRANSPORT TIEDOWNS. Ensure tiedown techniques in paragraph 8.10 are used for nuclear weapon ground support equipment.

12.15 ELECTRICAL SUBSYSTEMS AND EMR HAZARDS.

12.15.1 Aircraft and Missile Electrical Subsystems. See paragraph 12.4 for applicability.

12.15.2 General EMR Evaluation Requirements. These requirements outline responsibilities to ensure electromagnetic compatibility (EMC) of all military communications-electronics equipment, subsystems, and systems during conceptual, design, acquisition, and operational phases. The applicable parts of the following

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documents resulting from the DOD EMC Program are required to evaluate whether a system is invulnerable to any credible EMR environment throughout the STS: EMC Program Plan, EMC Control Plan, EMC/EMI Measurement Program Plan, and the EMC/EMI Measurement Test Report. That portion of evaluation data and results on EMR hazards to EEDs and semiconductor devices is required by the evaluation agency before the system is accepted.

12.15.3 Specific EMR/EMP Evaluation Requirements. The evaluation shall include the following:

a. Data attesting to the design adequacy of static electricity, EMC, external EMR, EMI, and lightning protection. System invulnerability to these potential hazards must be demonstrated, either by full scale tests, scale model tests, or by analyses.

b. EMR/EMP susceptibility data and results must verify the shielding effectiveness of the total system. Tests or analyses must demonstrate the total nuclear weapon (includes launch platform/missile/rocket and nuclear weapon/warhead) system's invulnerability to continuous wave (CW) and pulse fields through the entire STS, over the frequency range of 100 KHz to 40 GHz.

12.16 GENERAL EVALUATION OF TEST EQUIPMENT. The criteria in this section are to be applied to nuclear weapon system built-in test equipment, AMAC test equipment, release/launch test equipment, nuclear weapon/warhead testers, component testers (for racks, line replaceable units (LRU), and so forth), and general test equipment. The following requirements are in addition to good industrial standards normally applied to this type of equipment. The Navy agency evaluating the item shall establish specific criteria and give the rationale for this criteria.

a. Environmental tests shall be conducted according to paragraph 12.2.

b. Analyses. Minimum analyses shall consist of:

(1) Circuit analyses of tester operating with the circuits of the equipment to be tested.

(2) Failure mode and effect analysis of the test device. This analysis must verify that faults within the device will not degrade the nuclear safety of the equipment to be tested.

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(3) Tester interface with the weapon system. This analysis must verify the test concept.

c. Perform a fit and function demonstration to verify that both mechanical and electrical construction is compatible with the weapon system to be tested.

d. Verification. Demonstrate operations and procedures at field level and take appropriate corrective action in problem areas. Verify each applicable section of each procedural document.

e. Maintenance and Procedures. Procedures and analysis for maintenance and inspection of testers will be furnished by the procuring agency to assure that the integrity of the testers can be verified prior to use.

12.17 ANALYSIS OF AUTOMATIC NUCLEAR WEAPON SYSTEM SUSCEPTIBILITY TO ACCIDENTAL OR UNAUTHORIZED ACTUATON.

12.17.1 General. The objective of an accidental or unauthorized actuation analysis is to reveal facts about automatic (or in certain cases, automated) nuclear weapon systems design. This will establish the basis on which an overall determination may be made concerning the probability of such actuation. These facts may reveal design deficiencies in the system whereby either human actions, component failures, or combinations may negate the nuclear safety design safeguards. In preparing a Technical Nuclear Safety Analysis, the NWSSG must determine to what extent an automatic or semiautomatic nuclear weapon system is susceptible to unauthorized or accidental actuation.

12.17.2 Analysis of Unauthorized Actuation.

a. An analysis of the nuclear weapon system's susceptibility to unauthorized actuation shall be based on the following threats:

(1) One cognizant agent who has been authorized access to nuclear command control communications or who would have the opportunity to gain access to nuclear command and control communications security material. The cognizant agent is assumed to have detailed system knowledge and unrestrained access to the operational weapon system.

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(2) Any number of third party agents who are not authorized access to any part of the system, the classified information pertaining to the system, or communications associated with the system.

(3) As many tools as can be carried by one person. (It must be presumed, however, that once a person penetrates system security, a great number of tools could be at his disposal.) Each specific threat must be treated uniquely, insofar as establishing a complete threat and making assumptions.

b. The analysis shall identify those system components which can be tampered with or bypassed to contribute to or cause an unauthorized event and shall determine:

(1) The preparations to be taken and procedures necessary to contribute to or cause an unauthorized event at each point identified to be susceptible. The time required for one person to complete the preparatory steps shall be estimated.

(2) The tools and equipment necessary to accomplish the event.

(3) The characteristics of associated ground, missile, and airborne equipment, mechanical procedures, and other features which provide the ability to detect any attempt to penetrate the system.

(4) The time required to detect such an unauthorized act and the time and methods required to thwart the attempt.

c. From the information provided above, the methods of accomplishing an unauthorized event shall be arranged in calculated order of most probable to least probable and the principal reason for the order will be summarized.

12.17.3 Control of Unauthorized Actuation Analyses. Access of these analysis reports is rigidly controlled. Reports control is effected by assigning appropriate security classifications, limiting printing, and limiting access on a strict need-to-know basis.

12.17.4 Analysis of Accidental Actuation (Equipment Malfunction)

a. Determine the likelihood of occurrence for significant undesired events such as accidental nuclear detonation, release,

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unlock, or inadvertent programmed launch and accidental motor ignition using techniques such as a fault tree analysis. The analysis shall determine the critical path and identify equipment and components which could, by malfunction, result in the occurrence of such events.

b. Interlocks and other fail-safe features to reduce or limit the probability of an accidental event caused by malfunction shall be identified and described.

c. By use of simplified schematics, the number and location of component failures, occurring either in release sequence or other logical order, which would be required to activate the system shall be described.

12.17.5 Analysis of Accidental Actuation (Human Error).

a. Referring to those points of susceptibility previously identified, describe the general type, frequency, and degree of maintenance performed on the equipment. Designate these points as critical areas of maintenance.

b. Describe each maintenance or operational procedure that involves replacement, repair, or checkout of any component, module, drawer, or black box which can be identified as critical with respect to initiating an accidental actuation of the nuclear weapon system.

c. Identify and describe the general types, quantity, and complexity of tools and equipment used to perform maintenance on these components.

d. Identify and describe each design feature or condition of the equipment which will cause the system to fail-safe, if incorrect procedures are used by maintenance or operational personnel.

e. Identify and describe each design feature which, from a human engineering standpoint, is specifically designed to prevent human error.

f. From the information described above, arrange all maintenance activities in order of their calculated likelihood (highest to lowest) of causing an accidental event.

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12.18 SECURITY.

a. When evaluating any security system, evaluate the proposed security procedures. Also, consider the interaction of all systems applicable to security.

b. Evaluate all safeguards to ensure that the entire nuclear system is secure. Also, evaluate the physical security of any new or extensively modified nuclear system.

c. Demonstrate that the security system provides:

(1) Protection against undetected access.

(2) Protection against loss of critical secure data.

(3) Detection of tampering.

d. Documentation or data must be provided which assures:

(1) Reliability of the system.

(2) Procedures for operation of the security system.

(3) Quick reaction security plans associated with the system.

12.19 MULTIPLEXED MONITOR, CONTROL, AND RELEASE SYSTEM (MMCRS). Nuclear safety must be substantiated for all AMAC/REL systems incorporating multiplexing techniques. Safety analysis shall be conducted by the procurement/development agency and the evaluation agency to verify that no condition can degrade nuclear safety and that the safety criteria are met.

Preparing Activity:
Navy-AS

(Project No. NUOR NOO8)

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